Supplementary Material: Executable Evidence Management for Aircraft Inspection

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S1. Real-image source audit and detection experiment

This section records the real-image case in the main article. The source is Innovation Hangar v2, released version 1, mirrored in the aircraft-defect study's public author repository. The project and mirrored export identify CC BY 4.0. The complete source manifest records original paths, image and annotation hashes, representative selection, inferred source groups and partition membership. The redistributed representative images retain the released pixels and license credit. No generated photograph or model-generated defect image is used in this case.

S1.1. Source families, grouping and annotation conversion

The released 10,722 images comprise 9,651 training, 642 validation and 429 test files under the author's original split. Stripping Roboflow augmentation suffixes produces 4,161 filename families, but numeric filename variants also encode earlier rotations and flips. We group numeric variants by their base identifier and prefer variant zero, then the smallest available variant; for equivalent released copies we prefer validation or test before training. Timestamped captures share their complete calendar-day group. Flip/quarter-rotation invariant perceptual hashes propose additional matches at distance at most 10; normalized correlation at least 0.985 or at least 20 ORB/RANSAC inliers with fraction at least 0.65 accepts a merge. Five additional merges reduce 227 preliminary groups to 222. The retained 1,339 images have no exact decoded-pixel overlap between partitions. These checks establish the recorded source partition, not a complete recovery of unavailable aircraft identities.

A seeded whole-group split assigns 122/33/33/34 groups to training/development/calibration/test. Before fitting any detector, one whole-group swap ensures every available category occurs in training. No prediction was used in that choice. All original image identifiers and the swap remain in the manifest. Polygon annotations are converted to their axis-aligned enclosing boxes; released bounding boxes are retained. Image coordinates, class identifiers and original annotations remain recoverable from the manifest.

S1.2. Training, producer identity and measured environment

The initial fit runs 16 epochs at 320 pixels, batch 8, AdamW learning rate 0.001, with COCO-pretrained YOLO11n. Epoch 4 supplies the early checkpoint. The initial final checkpoint is retained separately. Development then permits up to 48 additional epochs at 480 pixels, no mosaic, batch 8, AdamW learning rate 0.001 and patience 16. The final model maximizes the trainer's development AP50–95 over the categories present in that partition: crack and dent. The other three categories have only three retained capture-day groups each, assigned to training, calibration and test; they have no fourth group available for a separate development partition. Its operational confidence maximizes development F1 among 0.05, 0.10, 0.25 and 0.50, breaking ties toward the higher threshold. Neither calibration nor test detections enter training, checkpoint selection or threshold choice. The exact optimization arguments, epoch histories, selected model hashes and measured process/elapsed times are archived.

The initial fit takes 384.1 s elapsed. The extended fit takes 2430.1 s elapsed and 9342.2 process CPU seconds. Development selects confidence 0.10; the selected final checkpoint and full development ranking

are archived. The extension executes 48 epochs; development selects additional epoch 43. Across all 3,524 outputs, measured prediction batch calls take 201.7 s elapsed and 613.2 process CPU seconds. These calls include model preprocessing, inference and postprocessing; model construction, image-file decoding, the explicit digital gain operation, annotation scoring, COCO evaluation and registry operations are outside this prediction timer.

Inference uses Ultralytics 8.3.232, PyTorch 2.8.0+cpu and OpenCV 5.0.0 on a four-core Linux CPU allocation. The extended fit and all inference explicitly use four Torch threads; the initial fit requested four threads but the library set eight, as retained in the thread audit. The four producers are early weights, final weights, final weights after gain 0.5, and final weights after gain 1.5. Gain multiplies decoded BGR pixels, rounds and clips to 0–255 without JPEG recompression. These are digital producer perturbations, not newly acquired illumination measurements. Each signature hashes model weights, gain, resolution, operational confidence, raw-candidate confidence 0.001, NMS IoU 0.7 maximum 100 detections, batch size 8, float32 CPU inference and the implementation library versions. Each stored output contains prediction arrays, input-pixel hashes and its producer identity. The new adapter registers two quantities in the existing typed compiler and calls the reference event-containment and card-authorization functions. A protocol cross-check applies this API to all 64 archived CAD cards and all 64 pairs of CAD event relations; it agrees with the original independent rule planner and does not add numerical observations. Metric implementation and source-manifest identities are checked before reuse; negative type checks reject changed population, producer, service or incompatible units and prevent a pooled card from covering a per-class event.

S1.3. All calibrated events and held-out outcomes

One complete inferred source group supplies one binomial trial. Pooled recall is matched defects divided by annotated defects; false positives are unmatched predictions divided by images in that group. Matching sorts predictions by confidence and assigns the best unused annotation of the same class at IoU at least 0.5. Categories without any annotation in a group contribute no denominator to pooled recall. Duplicate or wrong-class predictions remain false positives. The 16 events share allocation 0.05/16. The table includes every failure and every refused event; test data do not revise these cards.

Table S1: Every fixed real-image event. Calibration uses 33 groups and test uses 34 groups.

Producer	Request	Cal. failures	Upper	Test failures	Admitted
Early model	Triage	33	1.0000	33	No
Early model	Review	33	1.0000	33	No
Early model	Coverage	33	1.0000	33	No
Early model	Strict	33	1.0000	33	No
Updated model	Triage	28	0.9694	29	No
Updated model	Review	30	0.9911	32	No
Updated model	Coverage	33	1.0000	34	No
Updated model	Strict	33	1.0000	34	No
Gain 0.5	Triage	30	0.9911	29	No
Gain 0.5	Review	31	0.9975	31	No
Gain 0.5	Coverage	33	1.0000	33	No
Gain 0.5	Strict	33	1.0000	33	No
Gain 1.5	Triage	29	0.9815	30	No
Gain 1.5	Review	31	0.9975	33	No
Gain 1.5	Coverage	33	1.0000	34	No
Gain 1.5	Strict	33	1.0000	34	No

An unreserved query requires recall at least 0.60 and false positives per image at most 0.75. A compatible stronger allocated card can cover it; otherwise it is refused without a fresh unallocated inversion. The

descriptive calibration sensitivity uses 100 subsets of 24 out of the same 33 groups. These overlapping subsets are not 100 independent experiments. Likewise, the supplementary image-as-unit calculation pools a different weighting of observations and treats repeated frames as if independent; it is a diagnostic of that alternative analysis, not a matched confidence guarantee for the group event.

At allowances 0.10, 0.25 and 0.40, the admitted-context counts are 0, 0, 0. Exact and containment-plus-fallback support agree in every subset and allowance. At the primary allowance, exact renewal evaluates 528 calibration-group records and containment evaluates 528.

S1.4. Class support and the meaning of the event

Crack and dent occupy many independent filename groups. Missing head, paint loss and scratch are concentrated in one capture-day group per calibration/test partition. Their per-frame counts can support descriptive detection metrics; they do not establish class-specific population risks. With one independent positive group, even zero failures has unadjusted one-sided 95% upper bound 0.95. The fixed family reserves pooled-group events, not project-specific per-category inspection quotas. A per-category service is a different metric and cannot inherit a pooled-recall card.

Table S2: Updated-model class results on test images. Group counts measure source support separately from box counts.

Category	Positive groups	GT	TP	FP	Recall (%)
crack	19	231	35	308	15.2
dent	18	216	37	156	17.1
missing-head	1	103	57	87	55.3
paint-off	1	46	20	108	43.5
scratch	1	45	8	67	17.8

S1.5. Report transitions and preserved artifacts

Each test source group follows the fixed 12-operation sequence. The stored report payload identifies the exact observation images and prediction arrays, with no scoring labels included. The independent event-history implementation agrees with the registry snapshot on every issuance decision. The 408 operations include 68 accepted repairs, 0 issued reports and 0 repair checkpoints satisfying the current quality admission. Invalid bindings issued: 0. A valid repair changes the binding revision; statistical admission is checked independently. Replayed transactions cannot leave quarantine or roll back a current version.

The delivered archive contains every representative image, converted label, source mapping, early and final detector weights, initial-fit checkpoint, epoch history, model-selection record, raw detection array, score, event card and transition row. The default reproduction reruns inference from pixels with the archived weights, rechecks grouped counts and binomial inversion, then rebuilds figures and documents. It preserves original measured timings. All earlier CAD, rendering and recovery studies remain byte-identical and are retained below as supplementary evidence.

S2. Sequential change campaign and geometric image challenge

S2.1. Populations, fixed families and independent units

The principal campaign uses 256 fresh calibration scenes and 64 fresh test scenes from the four-layout skin population. All eight producer configurations execute on every calibration scene, giving 2,048 calibration outputs. Each test scene follows the same 16 operations; the resulting 1,024 decisions per implementation are repeated workflow observations of 64 independent sources. The 64 stress alternatives share those source seeds and form paired interventions. The geometric challenge separately contains 8 development scenes and 64 test scenes, each rendered under four conditions. Its 256 test images give 512 evaluations across the two providers.

The campaign fixes eight producers and eight request events, with $K = 64$, family error $\beta = 0.05$, per-event allocation $\alpha = 0.05/64$ and allowance $\delta = 0.15$. The family statement concerns scene-level request failures for these fixed functions. Shared scene inputs induce dependence between producer/event outcomes; the union bound permits that dependence. It does not convert the marginal allowance into a probability of zero failures over an entire 16-operation trajectory. The independent test assessment evaluates the producer/request contexts actually executed by the campaign.

Each operation selects a complete named configuration. The combined configurations explicitly change multiple parameters; individual update names do not accumulate patches from preceding operations. All methods receive the same numerical inputs, request thresholds, calibration allocation and authoritative records. The population mismatch operation executes a stress input and compares its explicit population identity with the calibration identity.

Table S3: Sampling units and repeated observations. Challenge conditions and provider evaluations remain paired within a source.

Set	Sources	Repeated observations	Seed range / structure
Campaign calibration	256	8 producers	202629100–202629355
Campaign test	64	16 operations	202630100–202630163
Paired stress	64	1 alternative/source	Same primary seeds
Challenge development	8	4 conditions	Families 0 and 1
Challenge test	64	4 conditions	Families 0, 1, 2 and 3

Table S4: Complete campaign configurations. Focal length, exposure and illumination are factors relative to the sampled baseline; decoder entries are contrast thresholds.

Producer	Focal	Exposure	Raster	Light	Decoder
Baseline	1	1	768×576	1	18
Focal	1.15	1	768×576	1	18
Exposure	1	1.5	768×576	1	18
Wide ROI	1	1	896×672	1	18
Lighting	1	1	768×576	0.45	18
Decoder	1	1	768×576	1	10
Exposure/light/decoder	1	1.5	768×576	0.45	10
Focal/ROI	1.15	1	896×672	1	18

S2.2. Complete calibration catalogue and executed test events

Table S5 reports every fixed-family calibration bound. 54 of 64 contexts satisfy the allowance. The held-out assessment covers 48 producer/request contexts from 6 executed nominal configurations, with 64 test scenes per context. A dash denotes a configuration not executed in the test campaign. Test entries are observed failure counts; the fixed-family confidence statement comes from calibration. The test check does not alter the plans. A plus sign denotes the relaxed version of the corresponding request.

Table S5: All 64 calibrated contexts and independent failure counts for the executed test contexts. Calibration uses per-event error 0.05/64.

Producer	Request	Cal. k/n	Cal. U	Admit	Test k/n
Baseline	Optical	0/256	0.027561	yes	0/64
Baseline	Optical+	0/256	0.027561	yes	0/64
Baseline	Acquisition	5/256	0.064119	yes	2/64
Baseline	Acquisition+	0/256	0.027561	yes	0/64
Baseline	Association	6/256	0.070218	yes	2/64
Baseline	Association+	4/256	0.057801	yes	1/64
Baseline	Complete	10/256	0.093151	yes	2/64
Baseline	Complete+	4/256	0.057801	yes	1/64
Focal	Optical	0/256	0.027561	yes	—
Focal	Optical+	0/256	0.027561	yes	—
Focal	Acquisition	15/256	0.119800	yes	—
Focal	Acquisition+	0/256	0.027561	yes	—
Focal	Association	10/256	0.093151	yes	—
Focal	Association+	1/256	0.036524	yes	—
Focal	Complete	17/256	0.130051	yes	—
Focal	Complete+	1/256	0.036524	yes	—
Exposure	Optical	3/256	0.051196	yes	1/64
Exposure	Optical+	0/256	0.027561	yes	0/64
Exposure	Acquisition	48/256	0.274712	no	8/64
Exposure	Acquisition+	4/256	0.057801	yes	2/64
Exposure	Association	13/256	0.109339	yes	4/64
Exposure	Association+	9/256	0.087590	yes	1/64
Exposure	Complete	52/256	0.292211	no	8/64
Exposure	Complete+	12/256	0.104016	yes	2/64
Wide ROI	Optical	0/256	0.027561	yes	—
Wide ROI	Optical+	0/256	0.027561	yes	—
Wide ROI	Acquisition	6/256	0.070218	yes	—
Wide ROI	Acquisition+	0/256	0.027561	yes	—
Wide ROI	Association	8/256	0.081927	yes	—
Wide ROI	Association+	5/256	0.064119	yes	—
Wide ROI	Complete	12/256	0.104016	yes	—
Wide ROI	Complete+	5/256	0.064119	yes	—
Lighting	Optical	0/256	0.027561	yes	0/64
Lighting	Optical+	0/256	0.027561	yes	0/64
Lighting	Acquisition	40/256	0.239107	no	8/64
Lighting	Acquisition+	5/256	0.064119	yes	1/64
Lighting	Association	54/256	0.300893	no	17/64
Lighting	Association+	29/256	0.188508	no	7/64
Lighting	Complete	57/256	0.313836	no	17/64
Lighting	Complete+	29/256	0.188508	no	7/64
Decoder	Optical	0/256	0.027561	yes	0/64
Decoder	Optical+	0/256	0.027561	yes	0/64
Decoder	Acquisition	4/256	0.057801	yes	2/64
Decoder	Acquisition+	0/256	0.027561	yes	0/64
Decoder	Association	5/256	0.064119	yes	2/64
Decoder	Association+	5/256	0.064119	yes	1/64
Decoder	Complete	9/256	0.087590	yes	3/64
Decoder	Complete+	5/256	0.064119	yes	1/64
Exposure/light/decoder	Optical	3/256	0.051196	yes	1/64
Exposure/light/decoder	Optical+	0/256	0.027561	yes	0/64
Exposure/light/decoder	Acquisition	63/256	0.339454	no	10/64
Exposure/light/decoder	Acquisition+	9/256	0.087590	yes	2/64
Exposure/light/decoder	Association	27/256	0.179049	no	5/64
Exposure/light/decoder	Association+	14/256	0.114598	yes	2/64
Exposure/light/decoder	Complete	66/256	0.352138	no	10/64
Exposure/light/decoder	Complete+	17/256	0.130051	yes	3/64
Focal/ROI	Optical	0/256	0.027561	yes	0/64
Focal/ROI	Optical+	0/256	0.027561	yes	0/64
Focal/ROI	Acquisition	9/256	0.087590	yes	4/64
Focal/ROI	Acquisition+	0/256	0.027561	yes	1/64

Producer	Request	Cal. k/n	Cal. U	Admit	Test k/n
Focal/ROI	Association	3/256	0.051196	yes	3/64
Focal/ROI	Association+	0/256	0.027561	yes	2/64
Focal/ROI	Complete	10/256	0.093151	yes	4/64
Focal/ROI	Complete+	0/256	0.027561	yes	2/64

S2.3. Campaign decisions, revision state and measured work

The output packet authenticates the actual RGB raster, decoded and predicted coordinates, fitted residual, matched/planned counts, source identity, population and producer. Physical mapping error and correct-address coverage remain offline scoring fields. The three incremental methods use the same persistent numerical DAG. Whole renewal discards its cache for ordinary request, producer and rollback evaluations and serves as the numerical cost reference. Fault, repair and replay operations validate stored outputs in every implementation.

The state and issuance columns in Table S6 agree across all four implementations. Eligibility denotes admission by the exact event together with a matching population; issuance also requires the current authenticated output and active state. A valid numerical rollback advances the registry revision, so a late output from a previous revision remains invalid.

Table S6: Complete 16-operation state sequence, aggregated over the independent test sources. All matched implementations give these decisions.

Step	Operation	Request	State	Rev.	Eligible	Issued
1	Baseline	Complete	active	0	64/64	64/64
2	Relax request	Complete+	active	1	64/64	64/64
3	Exposure	Complete+	active	2	64/64	64/64
4	Tighten request	Complete	active	3	0/64	0/64
5	Lighting	Optical	active	4	64/64	64/64
6	Decoder	Complete+	active	5	64/64	64/64
7	Combined update	Complete+	active	6	64/64	64/64
8	Camera update	Complete+	active	7	64/64	64/64
9	Stale attachment	Complete+	quarantined	7	64/64	0/64
10	Out-of-order repair	Complete+	quarantined	7	64/64	0/64
11	Valid repair	Complete+	active	8	64/64	64/64
12	Repair replay	Complete+	active	8	64/64	64/64
13	Numerical rollback	Complete+	active	9	64/64	64/64
14	Population mismatch	Complete+	active	9	0/64	0/64
15	Late delivery	Complete+	quarantined	9	64/64	0/64
16	Rollback repair	Complete+	active	10	64/64	64/64

The table abbreviations map to the four protocol methods as follows: Whole is Whole renewal; DAG is DAG exact + registry; Contract is Containment + repair, labelled Containment in the main figures; and Rules is Rules + registry. DAG uses exact evidence with the ordinary dependency cache and authoritative registry. Contract implements containment with validated repair; Rules uses independently implemented interval/provider rules and event-log authority. The complete-family catalogue and the 16-step demand catalogue describe separate cold-start demand patterns and are not added together.

Full-catalogue containment reduces calibration-record evaluations from 13,824 to 8,960, a 35.19% reduction relative to exact dependency caching. The separate 16-step demand reduces evaluations from 2,048 to 1,536, a 25% reduction. Every complete-family implementation supports the same 54 of 64 contexts.

Each implementation issues 704 of 1,024 operations and restores all 128 eligible recovery checkpoints, including the final recovery for all 64 test sources. No invalid binding is issued.

Table S7: Two independent catalogue demand patterns. Each catalogue is built once per population and reused across test sources. Counts are not multiplied by the number of image sources.

Method	Demand pattern	Cards	Record tests	CPU (ms)	Wall (ms)
Whole	16-step demand	9	2,304	2.121	2.124
Whole	Complete family	64	16,384	13.483	13.488
DAG	16-step demand	8	2,048	1.735	1.739
DAG	Complete family	54	13,824	9.577	9.580
Contract	16-step demand	6	1,536	1.123	1.125
Contract	Complete family	35	8,960	6.324	6.326
Rules	16-step demand	6	1,536	1.308	1.312
Rules	Complete family	35	8,960	6.327	6.329

Table S8: End-to-end campaign totals. Each method executes 1,024 workflow decisions. Numerical CPU is a component of event CPU; evidence-catalogue preparation is reported separately.

Method	Issued	Wrong binding	Repair commits	Numerical CPU (s)	Event CPU (s)	Wall (s)
Whole	704	0	128	192.354	193.617	218.024
DAG	704	0	128	78.549	78.773	88.192
Contract	704	0	128	79.006	79.328	90.951
Rules	704	0	128	79.379	79.687	91.876

Table S9: Actual provider calls over all campaign operations. Request changes and valid numerical rollback can reuse cached stage outputs.

Method	Geometry	Radiance	Sensor	Decode	Pose	Optics
Whole	640	640	640	640	640	640
DAG	192	256	384	448	448	256
Contract	192	256	384	448	448	256
Rules	192	256	384	448	448	256

Table S10: Per-operation work across all sources. Provider calls and event times aggregate the 64 sources; new cards and record tests belong to the single shared catalogue.

Step	Method	Provider calls	New cards	Record tests	CPU (s)	Wall (s)
1	Whole	384	1	256	18.9381	21.3876
1	DAG	384	1	256	18.9104	20.9969
1	Contract	384	1	256	19.1005	21.6866
1	Rules	384	1	256	18.7160	21.8655
2	Whole	384	1	256	18.4454	20.9273
2	DAG	0	1	256	0.0690	0.0745
2	Contract	0	0	0	0.0668	0.0720
2	Rules	0	0	0	0.0721	0.0818
3	Whole	384	1	256	18.6747	21.4200
3	DAG	256	1	256	3.5331	3.9809
3	Contract	256	1	256	3.4590	3.9189
3	Rules	256	1	256	3.5970	4.1071
4	Whole	384	1	256	18.7079	21.2900
4	DAG	0	1	256	0.0670	0.0670
4	Contract	0	1	256	0.0662	0.0662
4	Rules	0	1	256	0.0667	0.0668
5	Whole	384	1	256	18.6111	21.2576
5	DAG	256	1	256	7.8781	8.8089
5	Contract	256	0	0	7.9667	9.0906
5	Rules	256	0	0	8.0772	9.2928
6	Whole	384	1	256	18.7261	21.2705
6	DAG	128	1	256	2.3917	2.6704
6	Contract	128	1	256	2.3991	2.6582
6	Rules	128	1	256	2.3886	2.7371
7	Whole	384	1	256	18.6829	20.6956
7	DAG	192	1	256	3.4937	3.8949
7	Contract	192	1	256	3.4798	3.9903
7	Rules	192	1	256	3.4786	3.9884
8	Whole	384	1	256	24.8705	27.6911
8	DAG	384	1	256	24.2894	27.2656
8	Contract	384	1	256	25.1476	29.0300
8	Rules	384	1	256	25.0255	28.5834
9	Whole	0	0	0	0.0037	0.0037
9	DAG	0	0	0	0.0039	0.0040
9	Contract	0	0	0	0.0047	0.0047
9	Rules	0	0	0	0.0035	0.0036
10	Whole	0	0	0	0.0070	0.0119
10	DAG	0	0	0	0.0098	0.0099
10	Contract	0	0	0	0.0078	0.0079
10	Rules	0	0	0	0.0077	0.0078
11	Whole	0	0	0	0.0089	0.0089
11	DAG	0	0	0	0.0066	0.0067
11	Contract	0	0	0	0.0073	0.0073
11	Rules	0	0	0	0.0068	0.0068
12	Whole	0	0	0	0.0042	0.0042
12	DAG	0	0	0	0.0046	0.0046
12	Contract	0	0	0	0.0050	0.0051
12	Rules	0	0	0	0.0043	0.0044
13	Whole	384	1	256	18.8959	20.8877
13	DAG	0	0	0	0.0702	0.0788
13	Contract	0	0	0	0.0678	0.0731
13	Rules	0	0	0	0.0705	0.1051
14	Whole	384	0	0	19.0273	21.1540
14	DAG	384	0	0	18.0330	20.3161
14	Contract	384	0	0	17.5351	20.3210
14	Rules	384	0	0	18.1589	20.9871
15	Whole	0	0	0	0.0046	0.0047
15	DAG	0	0	0	0.0042	0.0043
15	Contract	0	0	0	0.0051	0.0052

Step	Method	Provider calls	New cards	Record tests	CPU (s)	Wall (s)
15	Rules	0	0	0	0.0044	0.0238
16	Whole	0	0	0	0.0088	0.0088
16	DAG	0	0	0	0.0085	0.0085
16	Contract	0	0	0	0.0094	0.0145
16	Rules	0	0	0	0.0090	0.0146

Timing rotates method order by source and uses two worker processes with single-threaded BLAS. Initial nominal-scene sampling, process startup and file export are excluded from event timing; stress-input sampling is included in its event. Event CPU includes executed providers, offline geometric scoring, payload hashing, and registry and consistency checks. Catalogue planning is measured separately. The paired timing bootstrap resamples complete source trajectories, keeping every operation and method paired. Its 10,000 replicates use seed 202630900. The DAG event-CPU saving relative to whole renewal is 59.31%, with percentile 95% interval 58.50–60.12%. These times describe the shared Linux execution host recorded in the machine-readable summary.

S2.4. Geometric scene construction and development-only provider selection

The geometric challenge renders 64 independently seeded test scenes, stratified equally across longitudinal lap joints, rounded access hatches, compound-curved diagonal joints and raised repair doublers. The 8 development scenes use only the first two families. Test scenes in these families have new seeds; diagonal joints and repair doublers are excluded from provider selection. Every source is rendered under nominal, raking-light, oblique-view and foreground-occlusion conditions. The occluders are scene geometry, and each condition preserves the source's planned target set.

Blender Cycles uses 768 by 576 pixels, at most 32 samples per pixel, adaptive threshold 0.02, and no denoising. Geometric sheet boundaries, sealant, fastener heads, wear, reflective materials and actual area lights contribute to image formation. The fixed AgX transform is applied consistently. The exported exterior-surface mesh contains the skin, panel and doubler surfaces and rims; it excludes fastener heads and foreground equipment.

The scene-level fastener-head radius is sampled from 3.0–3.8 mm, with an independent per-head multiplier of 0.94–1.06; head height is 0.45–1.0 mm. The nominal spacing parameter is 27–34 mm, with perimeter discretization adjusting individual spacings. Surface relief is 1.2 mm for the lap joint, 2.3 mm for the hatch, 1.8 mm for the diagonal joint and 2.4 mm for the repair doubler. Nominal CAD-layout coordinates receive 0.12-mm per-axis Gaussian jitter; physical target coordinates add independent 0.08-mm per-axis displacement in the surface plane. Nominal focal length is 1,320–1,480 pixels and camera z is 0.88–0.98 m above the CAD origin. These settings define the geometric challenge's patch scale.

Nominal CAD addresses and physically displaced targets are placed on the exported skin geometry. The prior adds independent Gaussian translation noise with 3-mm per-axis standard deviation and rotation noise with 0.001-rad per-axis standard deviation. Detection and pose estimation receive pixels, CAD locations and this prior. Visibility and true physical locations enter only scoring. Missing and occluded sites remain in the planned denominator.

The frozen local-contrast provider uses threshold 18. The alternative normalizes dark contrast by local intensity variation and uses one of six predeclared (z, a) pairs, with $z \in \{1.0, 1.6, 2.2\}$ and $a \in \{4, 8\}$. Development selection weights every family/condition cell equally, minimizes association-event failure, then maximizes coverage, with a lexical provider-name tie-break. The selected provider is `normalized_z1_a8`. Table S11 retains all development alternatives and the frozen comparator.

S2.5. All family–condition outcomes and observable screening

The challenge uses the Association request's three thresholds in the main requirement table, evaluated within its own image population. The association event fails if correct-address coverage is below 0.8, the maximum fitted residual exceeds 8 pixels, or maximum mapping error across all planned sites exceeds 8 mm. Correct correspondence requires the same target identity, a center within 8 pixels, and a target that is

Table S11: Development results averaged equally over family/condition cells. Only the six alternatives participate in selecting the replacement provider.

Provider	Quality failure (%)	Coverage (%)	Selected
normalized_z1_a8	28.12	93.62	yes
normalized_z1_a4	28.12	93.36	no
frozen_dark_contrast	31.25	94.41	baseline
normalized_z1.6_a4	46.88	75.51	no
normalized_z1.6_a8	46.88	75.24	no
normalized_z2.2_a4	81.25	46.19	no
normalized_z2.2_a8	81.25	46.19	no

present and visible in the rendered image. The observable screen uses matched/planned count at least 0.8 and maximum fitted residual at most 8 pixels. An accepted-but-failed image passes this screen and fails the association event.

Mapping casts each planned physical target's image ray from the estimated camera and intersects the known exterior CAD mesh. A missing ray intersection remains infinite and counts as a mapping failure. Reported quantiles use all images, including those infinite values, with the nearest-rank order statistic. Component-failure counts can overlap; their union is the association-event failure count. The archived field name is `quality_failure`. Screening and quality proportions use all images unless an accepted subset is explicitly specified.

Table S12: Every family/condition/provider quality summary. Coverage is in percent; mapping median and 95th percentile are in mm; CPU is median estimation time in ms. An asterisk marks a structure family excluded from development. Miss denotes images with at least one failed skin intersection.

Provider	Family	Condition	n	Coverage	Map med.	Map 95th	Miss	CPU
Frozen	Lap	Nominal	16	95.66	0.779	1.000	0	36.698
Frozen	Lap	Raking	16	85.07	0.733	1.216	0	36.412
Frozen	Lap	Oblique	16	95.31	1.024	1.513	0	36.362
Frozen	Lap	Occluded	16	96.35	0.756	0.976	0	37.064
Frozen	Hatch	Nominal	16	94.17	0.664	0.903	0	37.850
Frozen	Hatch	Raking	16	94.80	0.637	0.758	0	42.374
Frozen	Hatch	Oblique	16	92.99	1.020	1.308	0	38.967
Frozen	Hatch	Occluded	16	83.63	0.698	1.163	0	38.349
Frozen	Diagonal*	Nominal	16	95.60	0.636	0.927	0	41.477
Frozen	Diagonal*	Raking	16	93.75	0.715	1.331	0	40.981
Frozen	Diagonal*	Oblique	16	95.31	0.652	1.319	0	40.142
Frozen	Diagonal*	Occluded	16	89.49	0.631	0.978	0	39.527
Frozen	Doubler*	Nominal	16	94.58	0.623	0.890	0	43.510
Frozen	Doubler*	Raking	16	92.15	0.933	2.167	0	42.521
Frozen	Doubler*	Oblique	16	95.53	0.767	1.190	0	42.519
Frozen	Doubler*	Occluded	16	82.59	0.634	0.967	0	39.447
Selected	Lap	Nominal	16	96.70	0.917	2.537	0	78.933
Selected	Lap	Raking	16	87.15	2.247	3.025	0	78.854
Selected	Lap	Oblique	16	96.18	1.237	2.056	0	77.458
Selected	Lap	Occluded	16	96.70	0.916	2.549	0	78.166
Selected	Hatch	Nominal	16	95.39	0.773	1.570	0	80.575
Selected	Hatch	Raking	16	94.16	1.420	3.212	0	92.785
Selected	Hatch	Oblique	16	93.73	1.193	2.080	0	87.186
Selected	Hatch	Occluded	16	80.93	0.952	1.678	0	83.540
Selected	Diagonal*	Nominal	16	96.02	0.700	2.213	0	89.918
Selected	Diagonal*	Raking	16	92.05	1.402	2.770	0	81.746
Selected	Diagonal*	Oblique	16	96.45	0.907	1.538	0	83.035
Selected	Diagonal*	Occluded	16	90.48	0.727	2.382	0	89.127
Selected	Doubler*	Nominal	16	96.95	1.065	2.331	0	83.442
Selected	Doubler*	Raking	16	85.58	2.125	3.291	0	86.839

Provider	Family	Condition	n	Coverage	Map med.	Map 95th	Miss	CPU
Selected	Doubler*	Oblique	16	96.31	1.688	2.581	0	99.171
Selected	Doubler*	Occluded	16	82.96	1.570	2.434	0	80.283

Table S13: Every family/condition/provider failure count. Cov., Res. and Map are component failures, Any is their union, Accept is observable-screen acceptance, and A+F counts accepted-but-failed images. Each row uses the full stratum image count from Table S12.

Provider	Family	Condition	Cov.	Res.	Map	Any	Accept	A+F
Frozen	Lap	Nominal	0	0	0	0	16	0
Frozen	Lap	Raking	4	7	0	7	9	0
Frozen	Lap	Oblique	0	0	0	0	16	0
Frozen	Lap	Occluded	0	0	0	0	16	0
Frozen	Hatch	Nominal	1	2	0	2	14	0
Frozen	Hatch	Raking	0	5	0	5	11	0
Frozen	Hatch	Oblique	1	4	0	4	12	0
Frozen	Hatch	Occluded	3	12	0	12	4	0
Frozen	Diagonal*	Nominal	0	1	0	1	15	0
Frozen	Diagonal*	Raking	1	6	0	6	10	0
Frozen	Diagonal*	Oblique	0	1	0	1	15	0
Frozen	Diagonal*	Occluded	0	8	0	8	8	0
Frozen	Doubler*	Nominal	0	6	0	6	10	0
Frozen	Doubler*	Raking	1	9	0	9	7	0
Frozen	Doubler*	Oblique	0	6	0	6	10	0
Frozen	Doubler*	Occluded	4	15	0	15	1	0
Selected	Lap	Nominal	0	2	0	2	14	0
Selected	Lap	Raking	2	5	0	6	10	0
Selected	Lap	Oblique	0	2	0	2	14	0
Selected	Lap	Occluded	0	2	0	2	14	0
Selected	Hatch	Nominal	0	6	0	6	10	0
Selected	Hatch	Raking	1	4	0	4	12	0
Selected	Hatch	Oblique	1	8	0	8	8	0
Selected	Hatch	Occluded	7	10	0	12	4	0
Selected	Diagonal*	Nominal	0	7	0	7	9	0
Selected	Diagonal*	Raking	1	5	0	5	11	0
Selected	Diagonal*	Oblique	0	2	0	2	14	0
Selected	Diagonal*	Occluded	0	9	0	9	7	0
Selected	Doubler*	Nominal	0	7	0	7	9	0
Selected	Doubler*	Raking	4	11	0	11	5	0
Selected	Doubler*	Oblique	0	6	0	6	10	0
Selected	Doubler*	Occluded	3	15	0	15	1	0

S2.6. Paired comparisons and resampling hierarchy

The challenge bootstrap draws 4,000 replicates with seed 2026101805. Each replicate resamples independent scenes within each of the four fixed geometry families and retains all conditions and both providers for the selected scenes. The 2.5th and 97.5th percentiles give descriptive paired intervals. The provider contrast is selected minus frozen; a negative quality-failure difference favors the selected provider. Accepted-but-failed contrasts use the fraction of all images. These descriptive intervals summarize the stratified challenge and do not replace the campaign's finite-family calibration bounds.

S2.7. Deterministic presentation, checks and reproduction

Campaign examples use the first calibration seed and first two test seeds in ascending source order. Their full arrays, observable payloads and traces are archived. The geometric gallery uses the first test source of each family: 202640000, 202641000, 202642000, 202643000. The rendering protocol specifies the associated condition panels. Image selection follows source identity; every test success and failure remains in the result files.

Table S14: Paired provider differences and scene-stratified percentile intervals, in percentage points.

Condition	Outcome	Selected − frozen	95% interval
Nominal	Coverage	1.27	[0.56, 2.04]
Nominal	Quality failure	20.31	[10.94, 31.25]
Nominal	Accepted + failed	0.00	[0.00, 0.00]
Raking	Coverage	-1.71	[-4.68, 0.78]
Raking	Quality failure	-1.56	[-15.62, 12.50]
Raking	Accepted + failed	0.00	[0.00, 0.00]
Oblique	Coverage	0.88	[0.09, 1.71]
Oblique	Quality failure	10.94	[0.00, 21.88]
Oblique	Accepted + failed	0.00	[0.00, 0.00]
Occluded	Coverage	-0.25	[-1.06, 0.56]
Occluded	Quality failure	4.69	[-6.25, 17.19]
Occluded	Accepted + failed	0.00	[0.00, 0.00]

Table S15: Within-source changes from nominal imaging. Pass-to-fail and fail-to-pass count association-event transitions; coverage changes and intervals are percentage points.

Provider	Condition	Pairs	Pass → fail	Fail → pass	Δ coverage	95% interval
Frozen	Oblique	64	4	2	-0.22	[-1.12, 0.60]
Frozen	Occluded	64	26	0	-6.98	[-7.82, -6.18]
Frozen	Raking	64	22	4	-3.56	[-5.97, -1.28]
Selected	Oblique	64	4	8	-0.60	[-1.32, 0.03]
Selected	Occluded	64	18	2	-8.50	[-9.41, -7.66]
Selected	Raking	64	11	7	-6.53	[-10.06, -3.61]

Calibration and campaign tables retain pixel, observable-payload and numerical hashes. The challenge archive retains every raw image, scene description, exterior-surface mesh, fitted camera and correspondence array. The independent checkers recompute statistical bounds and state transitions, compare actual stage calls, reexecute stored numerical examples, and reprocess the challenge pixels. Geometric verification separately checks camera projection, true-pose surface intersection and an independent plane/barycentric intersection implementation on the first scene of each family.

From the manuscript directory, the default verification and reporting entry point is:

```
export A20_PYTHON=/path/to/paper/python
export A20_DETECTOR_PYTHON=/path/to/detector/python
bash reproduce_artifacts.sh
```

This entry point preserves the archived experiment records and uses isolated copies for report regeneration. A fresh numerical reproduction writes new measurements, timing records and selection metadata. Run the following execution commands only in a separate copy of the manuscript source tree. Within that separate copy, the campaign execution interface is:

```
export OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1
cd studies/lifecycle_campaign_20261018
python calibrate.py
python run.py
python analyze.py
python verify.py
```

Within the separate copy's challenge analysis directory, the processing interface is:

```
python check_legacy.py
```

```
python run.py select --input ../rendering/development
python run.py process --input ../rendering/test --split test
python summarize.py
python verify.py
```

These commands replace the copied calibration, campaign, development-selection and test-processing records. Selection reproduces the archived development-only choice; test processing checks the frozen implementation hashes. Use single-threaded BLAS for analysis. The renderer's executable recipe and frozen protocol reconstruct the raw images. The reporting command `python reporting/completion_supplement.py` rebuilds this section from completed result files and refuses missing final inputs. The following supplementary sections preserve earlier experiments under their original populations and allocations.

S3. Independent risk checks, validated repair and geometric rendering

S3.1. A complete numerical inspection example

This section records the complementary $K = 48$ single-update study, its validated recovery experiment and its original renderer-transfer study. The sequential main experiment uses a separate $K = 64$ family and fresh source blocks. The example for this complementary study is the smallest seed in its held-out controlled population, 202619100. Its selection does not depend on image quality. Baseline and exposure outputs both satisfy every defined quality predicate for that individual scene. The population complete-service bound changes from 0.096886686 to 0.272263700, while the exposure producer's relaxed exact-event bound is 0.080307334. Geometric sampling remains 0.857746217 mm/pixel and blur changes from 1.049798653 to 1.574697979 pixels. The optical request includes blur, so its original card is renewed even though sampling is unchanged. No standalone sampling-only card is claimed.

The source-recovery trace retains the same capture identity and binding. Validated repair advances its revision to 2 and restores the authenticated exposure output. Relaxed complete and optical requests resume; strict complete service remains inadmissible because its population bound exceeds 0.15. A duplicate repair command is rejected without modifying the successful prior commit. Machine-readable values and all 39 request/step combinations for this source are in `studies/decision_case_20261018`.

S3.2. Risk selection, issuance and test-set interpretation

The main proposition separates two random quantities. Calibration data C determine upper bounds and the selected fixed producer/event. An independent future scene X determines whether that event fails. With probability at least $1 - \beta$ over C, every admitted calibration-dependent selection satisfies its fresh-scene risk allowance, provided the event inclusion and population/function assumptions hold. Paired producers on the same scene can be dependent; the finite-family union bound does not require independence between event rows.

The fixed family includes every exact fallback event. A new relaxed event can inherit an already allocated anchor's bound through pointwise inclusion. Constructing a new exact-event bound outside the reserved family instead requires its own confidence allocation. Selection in the main proposition is committed before observing the new scene; choosing a different event after seeing its outcome is a different policy.

A later attachment gate I gives the joint bound $\Pr(E \cap \{I = 1\} \mid C) \leq \delta$. A bound conditional on issuance additionally requires an appropriate independence condition or calibration of that complete selection policy. For example, let X be Bernoulli with probability 0.1, $E = \{X = 1\}$ and $I = X$. Then $\Pr(E) = \Pr(E \cap \{I = 1\}) = 0.1$, whereas $\Pr(E \mid I = 1) = 1$. Output integrity checks do not by themselves provide a new conditional-risk theorem.

The held-out risk check evaluates all 48 fixed programs on all 256 test scenes, with a separate allocation 0.05/48. It is performed after the original plans are fixed and does not select new thresholds or plans. All 41 calibration-admitted programs have held-out upper bounds at most 0.074568224. This is a separate 95% simultaneous test-family statement. Combining two separate 95% calibration and test statements does not create a joint 95% statement; a direct union bound would give at least 90% for their intersection.

Table S16: Allowance sensitivity with the same fixed 48-event family. Complete methods have identical support. Every constructed card evaluates 256 stored calibration records.

Allowance	Support	Whole cards	DAG cards	Contract/rules cards
0.02	0/48	48	42	42
0.05	18/48	48	42	38
0.10	40/48	48	42	27
0.15	41/48	48	42	26
0.20	43/48	48	42	26

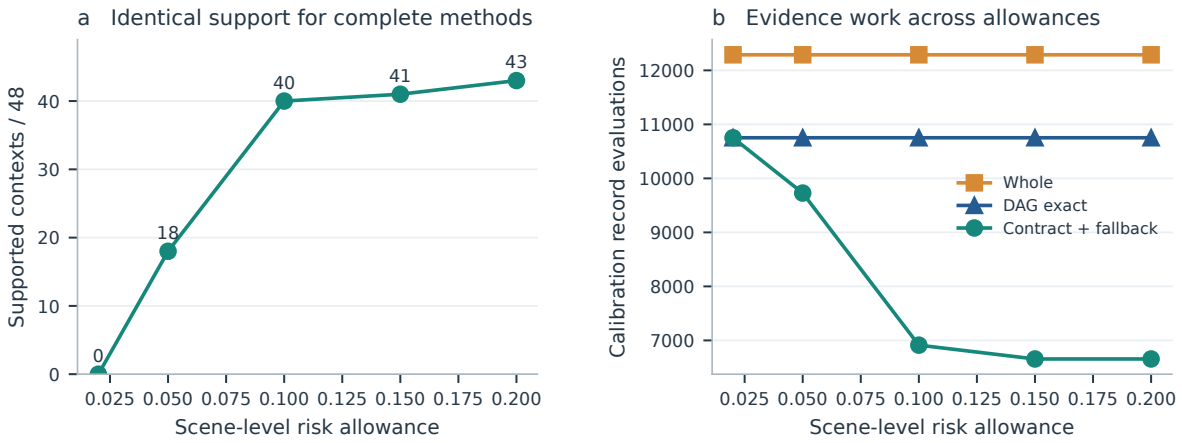


Figure S1: Sensitivity to the scene-level quality allowance. Support is identical across complete methods. Construction includes refused events and failed anchors.

Table S17: Independent test-family bounds. Each row evaluates 256 held-out scenes; selected refers to the original calibration decision at allowance 0.15.

Producer	Request	Test failures	Test upper	Selected
baseline	optical	0	0.026467	yes
baseline	optical relaxed	0	0.026467	yes
baseline	acquisition	2	0.042897	yes
baseline	acquisition relaxed	0	0.026467	yes
baseline	association	3	0.049839	yes
baseline	association relaxed	2	0.042897	yes
baseline	complete	4	0.056382	yes
baseline	complete relaxed	2	0.042897	yes
focal	optical	0	0.026467	yes
focal	optical relaxed	0	0.026467	yes
focal	acquisition	5	0.062643	yes
focal	acquisition relaxed	0	0.026467	yes
focal	association	4	0.056382	yes
focal	association relaxed	2	0.042897	yes
focal	complete	7	0.074568	yes
focal	complete relaxed	2	0.042897	yes
exposure	optical	1	0.035323	yes
exposure	optical relaxed	0	0.026467	yes
exposure	acquisition	27	0.176898	no
exposure	acquisition relaxed	1	0.035323	yes
exposure	association	3	0.049839	yes
exposure	association relaxed	1	0.035323	yes
exposure	complete	29	0.186320	no
exposure	complete relaxed	2	0.042897	yes
wide roi	optical	0	0.026467	yes
wide roi	optical relaxed	0	0.026467	yes
wide roi	acquisition	2	0.042897	yes
wide roi	acquisition relaxed	0	0.026467	yes

Producer	Request	Test failures	Test upper	Selected
wide roi	association	3	0.049839	yes
wide roi	association relaxed	1	0.035323	yes
wide roi	complete	4	0.056382	yes
wide roi	complete relaxed	1	0.035323	yes
lighting	optical	0	0.026467	yes
lighting	optical relaxed	0	0.026467	yes
lighting	acquisition	35	0.214090	no
lighting	acquisition relaxed	7	0.074568	yes
lighting	association	48	0.272264	no
lighting	association relaxed	26	0.172152	no
lighting	complete	49	0.276646	no
lighting	complete relaxed	26	0.172152	no
decoder	optical	0	0.026467	yes
decoder	optical relaxed	0	0.026467	yes
decoder	acquisition	1	0.035323	yes
decoder	acquisition relaxed	0	0.026467	yes
decoder	association	1	0.035323	yes
decoder	association relaxed	1	0.035323	yes
decoder	complete	2	0.042897	yes
decoder	complete relaxed	1	0.035323	yes

S3.3. Repair model, trust boundary and independent checks

The authoritative record identifies binding, capture, revision, producer, payload digest and calibration lineage. An output envelope references the immutable numerical archive by its SHA-256, phase, seed and producer variant. Its observable payload includes fitted pixel residual and matched/planned counts. Coverage, physical mapping and optical quality fields are used only for independent scoring and never for runtime authorization.

The two numerical-origin faults attach either the same source's old baseline record after an exposure update or the next source's exposure record. All substituted records differ in their observable numerical values. The oracle checks the attached archive identity and intended source/producer, active registered binding and risk plan independently of the candidate gate. Protocol availability is counted separately from wrong numerical origin: a refusal after an invalid attempt is a quarantine rule, not evidence that the corrected numerical output itself has become wrong.

After quarantine, repair presents an authority-issued certificate with an expected revision and a strictly next revision. It binds source, active producer, trusted payload digest and calibration identity. Missing evidence, a replayed prior certificate, wrong payload, wrong producer and inactive calibration fail validation. A renamed binding lacks a registered source relationship and fails as well. A valid commit restores the binding at revision 2; a duplicate commit is rejected transactionally while preserving the already active state. The audit log retains the original invalid attempt.

The experiment assumes that the authority, immutable archive and calibration registry are trusted. Content hashes check integrity; the authority's controlled transaction interface establishes trust. The prototype uses an in-process trusted registry rather than a deployed public-key or distributed consensus service. Compromising that authority is outside the evaluated fault model. A production implementation can preserve these semantics with its authenticated storage and transaction mechanism.

There are 256 source observations, eight requests, two independent fault scopes and 13 transitions, producing 53,248 decision rows per implementation. Repeated workflow operations are not independent image samples. Of the 4,096 fault scenarios, 3,072 have a supported exposure request and are eligible for post-repair issuance. The 4,096 valid repair commits also include statistically refused requests; a correct binding alone does not authorize them. The stronger three implementations restore all 3,072 eligible scenarios and agree at every step. The permanent latch restores none. No additional image rendering or solving is needed when the already computed trusted output is available.

The authoritative-snapshot baseline reads the current trusted state. The event-sourced baseline independently folds the event log. Their agreement with the contract attributes the result to common provenance and repair semantics, not to a unique representation. The metadata-only ablation trusts self-declared producer information and supplies a fault control. Independent tests mutate authentic certificates, attempt rollback and skipped revisions, alter payloads, substitute source aliases, replay a completed transaction and remove latent truth fields. Reproduction and invariant reports are archived with the study.

S3.4. Cycles geometry, materials and processing

The renderer-transfer study uses Blender 4.3.2 Cycles on the same Linux host. The surface comprises two curved triangulated panels with a 0.25-mm lap offset, geometric sealant joints, shallow domed fastener heads, countersink rings and abrasion meshes. Three manufactured layouts are sampled independently with curvature, camera pose, missing heads, roughness and painted/metallic appearance. These are synthetic surface CAD models. Every scene is reconstructed from its integer seed.

The held-out seeds are 202618800–202618823. Development uses 202618701, 202618702, 202618703 and 202618706. Initial pilot brightness and surface appearance were adjusted before freezing the held-out distribution. The detector and estimator are unchanged throughout. Each held-out source renders a baseline, +1.3-stop radiometric gain and 0.1-times area-light/world-radiance configuration. Resolution is 768×576, maximum Cycles samples per pixel is 32, adaptive sampling threshold is 0.01, and denoising is disabled. The AgX display transform is fixed. The experiment has a static pinhole camera; gain does not model shutter duration or motion blur.

The original pixel detector receives only the rendered RGB image. CAD points and a perturbed pose prior support assignment and refinement. Blender's independent camera coordinates are compared with the numerical projector to a tolerance of 0.001 pixel. Mapping is evaluated using the estimated transform at every planned physical-site ray, including missing heads, against the original analytical base-skin model. The lap offset contributes a known surface-model discrepancy. Fitted residual includes every assigned correspondence. Thus mapping measures the recovered geometry, while coverage measures correct extracted-site identities.

The association event uses inherited limits: coverage at least 0.8, fitted residual at most eight pixels and maximum mapping error at most 8 mm. The archive's auxiliary acquisition-failure counter checks coverage alone and is not used as the complete acquisition event. All 72 image results enter the summaries below; none is selected out after fitting. The small transfer study has its own population identity and does not borrow the controlled renderer's calibrated probability bound.

Table S18: Every held-out geometric-rendering result. Coverage is in percent, residual in pixels and mapping in mm. Gain denotes +1.3 stops.

Seed	Variant	Coverage	Residual	Mapping	Matched
202618800	baseline	91.2	2.079	0.883	31
202618800	gain	91.2	3.756	1.004	31
202618800	lighting	91.2	4.611	0.802	31
202618801	baseline	67.6	1.281	0.665	23
202618801	gain	88.2	1.543	0.765	30
202618801	lighting	79.4	4.305	0.739	27
202618802	baseline	88.2	1.240	0.960	30
202618802	gain	91.2	1.611	1.045	31
202618802	lighting	76.5	1.386	0.860	26
202618803	baseline	90.9	0.929	0.729	40
202618803	gain	84.1	1.031	0.853	37
202618803	lighting	90.9	1.049	0.657	40
202618804	baseline	87.5	28.554	0.532	37
202618804	gain	87.5	28.486	0.650	37
202618804	lighting	5.0	12.208	7.137	2
202618805	baseline	90.0	16.839	0.545	37

Seed	Variant	Coverage	Residual	Mapping	Matched
202618805	gain	92.5	12.312	0.751	38
202618805	lighting	67.5	15.266	1.861	30
202618806	baseline	85.0	1.651	0.816	34
202618806	gain	85.0	2.107	0.973	34
202618806	lighting	12.5	3.767	2.465	5
202618807	baseline	79.4	1.549	0.861	27
202618807	gain	91.2	1.631	0.786	31
202618807	lighting	91.2	1.353	0.638	31
202618808	baseline	91.2	1.222	0.722	31
202618808	gain	91.2	1.520	0.828	31
202618808	lighting	91.2	4.534	0.564	31
202618809	baseline	95.0	1.445	0.839	38
202618809	gain	90.0	35.529	0.967	38
202618809	lighting	85.0	4.488	0.717	34
202618810	baseline	82.5	2.028	0.922	33
202618810	gain	90.0	22.442	1.108	37
202618810	lighting	92.5	4.040	1.674	37
202618811	baseline	68.2	0.999	0.594	30
202618811	gain	86.4	1.117	0.666	38
202618811	lighting	68.2	20.176	0.591	31
202618812	baseline	65.9	1.193	0.480	29
202618812	gain	88.6	4.468	0.502	39
202618812	lighting	54.5	20.531	0.709	26
202618813	baseline	90.9	1.128	0.767	40
202618813	gain	90.9	1.379	0.880	40
202618813	lighting	93.2	19.797	1.256	42
202618814	baseline	82.5	1.028	0.571	33
202618814	gain	82.5	1.435	0.683	33
202618814	lighting	87.5	4.994	2.700	35
202618815	baseline	95.0	1.457	0.720	38
202618815	gain	95.0	8.832	0.809	38
202618815	lighting	97.5	4.971	2.319	39
202618816	baseline	79.4	1.374	0.447	27
202618816	gain	82.4	1.999	1.039	28
202618816	lighting	91.2	4.021	0.667	31
202618817	baseline	86.4	1.149	0.783	38
202618817	gain	90.9	1.518	0.946	40
202618817	lighting	93.2	3.912	0.722	41
202618818	baseline	79.4	2.870	0.883	27
202618818	gain	91.2	4.605	1.204	31
202618818	lighting	50.0	6.646	2.661	17
202618819	baseline	90.0	1.655	0.908	36
202618819	gain	95.0	2.131	1.093	38
202618819	lighting	95.0	1.552	0.816	38
202618820	baseline	81.8	1.395	0.637	36
202618820	gain	88.6	1.742	0.775	39
202618820	lighting	90.9	23.388	2.605	41
202618821	baseline	84.1	0.843	0.789	37
202618821	gain	86.4	1.022	0.932	38
202618821	lighting	81.8	0.777	0.690	36
202618822	baseline	77.5	1.490	0.859	31
202618822	gain	95.0	1.925	1.108	38
202618822	lighting	95.0	4.602	0.742	38
202618823	baseline	94.1	1.552	0.725	32
202618823	gain	94.1	1.823	0.876	32

Seed	Variant	Coverage	Residual	Mapping	Matched
202618823	lighting	94.1	4.227	0.780	32

S3.5. Reproduction and archived presentation

The source package preserves numerical inputs, original rendered PNGs, scene metadata, correspondence arrays, source hashes and an executable scene recipe. A representative Blender scene is included. Deterministic scripts regenerate the main and supplementary figures, which are supplied as publication PDFs. Duplicate build PDFs and generated preview PNG/SVG files may be omitted from the ZIP; their source scripts regenerate them from the preserved observations. This packaging does not discard experimental pixels or numerical records.

The earlier controlled image and numerical studies follow below with their original identities and allocations. Their permanent-latch conformance counts document the earlier protocol; the sequential campaign in the main article extends the validated-repair model examined here. Earlier numerical observations are preserved byte for byte.

S4. Controlled image population and archived lifecycle comparisons

S4.1. Population, producer and independent sampling

The complementary single-update study is the four-layout aircraft-skin simulation in `studies/lifecycle_20261018`. It contains 256 calibration scenes, 256 held-out scenes, 128 separate stress scenes, 16 timing scenes and 64 regression scenes. Their disjoint seed ranges begin at 202618100, 202619100, 202620100, 202621100 and 202622100, respectively. Four development scenes use seeds beginning at 202618000. A complete scene is the sampling unit. Producer updates and request labels are repeated measurements of that scene.

The six main producer configurations execute on every calibration and held-out scene, giving 3,072 outputs. The 128 stress scenes execute the baseline producer. The four layout families are sampled independently with equal probability. Parallel rows have 36 sites, curved rows have 28, staggered layouts have 35 and access-panel perimeters have 28. The geometry remains an analytic quadratic patch with independently sampled planar, singly curved or doubly curved family. Principal radii, camera pose and optical ranges are inherited from the controlled study below; the fastener layout, missingness, appearance and CAD discrepancies are newly generated.

Layout spacing is sampled between 28 and 40 mm where a layout uses a spacing parameter. The layout rotates by a uniform angle in $[-0.3, 0.3]$ radians and each nominal site receives 0.25-mm coordinate jitter. Physical site coordinates then differ from their nominal CAD addresses through a shared two-dimensional normal displacement with component standard deviation 0.4 mm and independent local displacements of 0.15 mm. Site heights follow the sampled physical surface. Head radii vary independently between 2.0 and 3.4 mm. A site is missing with probability 0.025.

The renderer intersects camera rays with the surface and uses nearest-site distances to place flush heads and a countersink normal map. Painted and metallic surfaces have separate diffuse/specular coefficients, with roughness sampled in $[0.22, 0.48]$, brushed variation, panel joints, a spatial stain, shallow scratch marks and independent intensity noise. An opaque right-edge region covers 0–10% of raster area. Exposure filtering and intensity noise produce the observed raster. This appearance model supplies controlled geometric and photometric variation; it is not a measured aircraft reflectance model.

The stress population raises missing-site probability to 0.14, maximum occluded area to 35%, shared CAD-displacement standard deviation to 2 mm and local discrepancy to 0.6 mm. It also uses 7-mm prior-translation component standard deviation, compared with 3 mm nominally, and intensity noise up to 8 rather than 4 units. It is a separate population and supplies no calibration records for the nominal cards.

S4.2. Pixel estimation and scoring

The nominal decoder thresholds Gaussian local contrast at 18 and retains connected components with 5–240 pixels. Intensity-weighted centers are assigned to CAD predictions from the pose prior with a 30-pixel gate. At least six assignments trigger six-degree-of-freedom Huber refinement. The objective scales are 0.5 pixel, 3 mm and 0.001 rad, with at most 60 function evaluations and 10^{-8} termination tolerances. The decoder update lowers the contrast threshold to 10.

The estimator sees nominal CAD addresses, decoded pixels and the supplied prior. Scoring then uses physical site coordinates. Correct-address coverage requires an assigned center to have the same nearest physical identity, to lie within eight pixels of that identity and to correspond to a present fastener. Coverage divides by all planned sites. Reprojection is the maximum residual over every assigned correspondence, including incorrectly assigned ones. Mapping casts rays through every true physical image location using the estimated pose, intersects the CAD surface and takes the maximum displacement across all planned sites. No ground-truth correspondence filtering removes difficult residuals from the new study. An empty assigned set has infinite reprojection error.

S4.3. Request family and compiler

The eight request intervals are copied unchanged from the controlled 20261014 surface protocol before the new population runs. The existing unit-aware compiler converts millimetres to metres and checks canonical event identities and provider projections. The new code asserts that every compiled bound equals the inherited

Table S19: New scene counts by layout. Geometry and appearance are sampled independently of this layout draw.

Layout	Calibration	Held out	Stress
Parallel rows	63	62	36
Curved rows	66	73	31
Staggered	56	60	30
Access perimeter	71	61	31

specification. Calibration uses scene-level union failure, $n = 256$, family $K = 48$, confidence error 0.05 and allowance 0.15. The exact and reuse methods retain this same allocation. Matching a numerical-stage cache does not itself supply a statistical statement.

Table S20: Request limits for both surface studies. Optical sampling and mapping limits are shown in mm; blur and residual limits in pixels.

Request	GSD $\leq$	Blur $\leq$	Coverage $\geq$	Residual $\leq$	Mapping $\leq$
optical	1.2	3	—	—	—
optical relaxed	1.5	4	—	—	—
acquisition	1	2	0.8	—	—
acquisition relaxed	1.2	3	0.6	—	—
association	—	—	0.8	8	8
association relaxed	—	—	0.6	12	12
complete	1	2	0.8	8	8
complete relaxed	1.2	3	0.6	12	12

S4.4. Stage cache, evidence construction and timing

The implemented graph has geometry, radiance, sensor, decoding, pose and optical nodes. Radiance depends on geometry; sensor pixels depend on radiance; decoding depends on sensor pixels; pose depends on decoded pixels and the registered CAD/prior parameters. Explicit optical quantities have a separate branch. Every node key contains its stage identity, direct parameter values and parent keys. Stage implementations are fixed within the experiment and are versioned by the archived source manifest. Numerical parameters and stage outputs determine reuse.

Each timing case starts with the same fully evaluated baseline cache. Whole renewal discards those stage values and executes the complete updated producer. DAG exact, contract and rules paths each start from a fresh copy of that baseline cache and evaluate the requested output stages. A scalar event check is included. Required numerical values are compared across all four executed methods to relative tolerance 10^{-10} and absolute tolerance 10^{-9}. Method order rotates by scene, update and request. All 16 inputs execute five updates and eight requests, giving 640 cases per method and 2,560 timed executions in total. These timings do not count repeated requests as independent scene samples.

The timer records process CPU and elapsed time around the actual requested computation. Population generation uses two single-thread workers and the timing stream one; these jobs share the host. The reported paired bootstrap resamples the 16 scene-level workload sums together across methods, with 10,000 resamples and seed 20261818. It summarizes this host/workload measurement, rather than hardware-independent real-time latency.

Catalogue timing uses all stored calibration quantities and eleven repetitions. Static provider projections and interval relations are compiled and cached outside repeated builds, equally for the methods using those relations. Event evaluations, bound inversion and route selection remain inside the timer. Uncompiled catalogue timings are also retained in the package. No confidence family or request threshold changes during this implementation optimization.

For this complementary $K = 48$ single-update experiment, whole renewal constructs 48 cards and evaluates 12,288 calibration records; the dependency-aware exact cache constructs 42 cards and evaluates

10,752 records. Containment with exact fallback and the independent rules each construct 26 cards and evaluate 6,656 records, a 38.10% reduction relative to exact caching. All four complete methods support the same 41 contexts. These counts belong to the archived 48-event population, separately from the main article's 64-event sequential campaign.

Table S21: Executed source renewal across all 640 cases per method.

Method	Process CPU (s)	Elapsed sum (s)
Whole renewal	183.992576	184.227523
DAG exact cache	80.485837	80.661254
Contract + fallback	80.278759	80.476726
Rules + containment	79.756769	79.836443

S4.5. Component-update regression and attachment traces

The 64-input registration experiment compares each clean update and a conditional coupled update with the same baseline source. ROI additionally changes focal length by 8% when the camera's lateral coordinate is positive. Reduced illumination additionally multiplies exposure by 1.8 on painted surfaces. The decoder update additionally changes focal length by 8% for doubly curved surfaces. These are specified coupling fixtures. A quantity change beyond 10^{-10} absolute/relative tolerance vetoes registration when the declared update leaves that quantity's provider projection unchanged. Clean updates have no undeclared differences. A successful finite regression suite cannot establish functional identity over all possible inputs; it can expose a counterexample to that interface.

For each of the 256 held-out image sources and eight requests, the lifecycle study registers and checks numerical quality-output payloads generated by the new producer. Their contents are coverage, reprojection and mapping error, together with source and producer identities. The payload digest is a numerical-record integrity check. Source versions advance monotonically. Three operating stages evaluate baseline, increased exposure and restoration; a final independent successor tests new-scope recovery. Separate stale-version, altered-payload and wrong-producer scopes each include the invalid attempt and a reconstructed-plan retry. Thus each context has ten decisions per implementation/ablation. The four variants produce 81,920 protocol decisions over the same 256 held-out sources; this count is not an additional statistical sample size.

This archived experiment evaluates the original permanent-latch protocol. A refused statistical event leaves attachment history intact; an invalid version, producer or payload latches its fault scope. The snapshot ablation checks only the current attachment, the stateless ablation omits version/integrity checks, and the payload ablation retains version/history while omitting the digest comparison. The counts measure conformance to that earlier irreversible rule. The complementary authenticated-repair experiment in Section S3 compares strong registry and event-log baselines; the main article extends those semantics to the sequential campaign. The earlier figures and records remain archived for reproducibility.

Table S22: Scene-level held-out and stress quality. Mapping errors are maxima across all physical sites before the distribution summary.

Population	Producer	Coverage (%)	Residual (px)	Map median (mm)	Map p95 (mm)
stress	baseline	72.69	2.830	2.712	13.095
test	baseline	96.14	0.639	0.502	1.052
test	decoder	96.27	0.636	0.501	1.045
test	exposure	95.98	0.642	0.498	1.045
test	focal	96.21	0.718	0.489	1.069
test	lighting	91.76	0.686	0.529	1.262
test	wide roi	96.15	0.648	0.500	1.053

S4.6. All 48 risk cards in the new population

Table S23: New-population calibration counts and independent test failures. Every row has 256 calibration and 256 test inputs.

Producer	Request	Calibration k	Upper U	Test failures	Admit
baseline	optical	0	0.026467	0	yes
baseline	optical relaxed	0	0.026467	0	yes
baseline	acquisition	6	0.068690	2	yes
baseline	acquisition relaxed	0	0.026467	0	yes
baseline	association	8	0.080307	3	yes
baseline	association relaxed	5	0.062643	2	yes
baseline	complete	11	0.096887	4	yes
baseline	complete relaxed	5	0.062643	2	yes
focal	optical	0	0.026467	0	yes
focal	optical relaxed	0	0.026467	0	yes
focal	acquisition	10	0.091452	5	yes
focal	acquisition relaxed	0	0.026467	0	yes
focal	association	6	0.068690	4	yes
focal	association relaxed	5	0.062643	2	yes
focal	complete	14	0.112762	7	yes
focal	complete relaxed	5	0.062643	2	yes
exposure	optical	3	0.049839	1	yes
exposure	optical relaxed	0	0.026467	0	yes
exposure	acquisition	43	0.250167	27	no
exposure	acquisition relaxed	3	0.049839	1	yes
exposure	association	11	0.096887	3	yes
exposure	association relaxed	5	0.062643	1	yes
exposure	complete	48	0.272264	29	no
exposure	complete relaxed	8	0.080307	2	yes
wide roi	optical	0	0.026467	0	yes
wide roi	optical relaxed	0	0.026467	0	yes
wide roi	acquisition	5	0.062643	2	yes
wide roi	acquisition relaxed	0	0.026467	0	yes
wide roi	association	6	0.068690	3	yes
wide roi	association relaxed	4	0.056382	1	yes
wide roi	complete	9	0.085930	4	yes
wide roi	complete relaxed	4	0.056382	1	yes
lighting	optical	0	0.026467	0	yes
lighting	optical relaxed	0	0.026467	0	yes
lighting	acquisition	35	0.214090	35	no
lighting	acquisition relaxed	6	0.068690	7	yes
lighting	association	45	0.259044	48	no
lighting	association relaxed	25	0.167381	26	no
lighting	complete	47	0.267869	49	no
lighting	complete relaxed	25	0.167381	26	no
decoder	optical	0	0.026467	0	yes
decoder	optical relaxed	0	0.026467	0	yes
decoder	acquisition	3	0.049839	1	yes
decoder	acquisition relaxed	0	0.026467	0	yes
decoder	association	5	0.062643	1	yes
decoder	association relaxed	5	0.062643	1	yes
decoder	complete	8	0.080307	2	yes
decoder	complete relaxed	5	0.062643	1	yes

S4.7. Reproduction and provenance

The package includes the renderer, stage cache, separate interval/rule implementations, unit compiler, all observations and configurations, per-case timing and stage counts, all cards, numerical regression pairs and full lifecycle decisions. Source checksums identify the executed versions; implementation notes record compiled-metadata optimization and clarification of attachment scopes. These changes leave image generation, sampling seeds, request thresholds and numerical observations unchanged. Independent re-execution checks the published numerical observations and example pixels. The complementary studies below retain their original inputs, allocations, clocks and identities.

S5. Controlled regular-grid CAD-surface benchmark

S5.1. Scene generation and independent inputs

The controlled regular-grid study uses an analytic surface producer, separate from the earlier diagnostic-fiducial producer. It draws 384 complete scenes, with 192 calibration and 192 test inputs. The two seed ranges begin at 202615100 and 202616100. Every scene executes the baseline and five registered interventions, giving 2,304 producer/input evaluations. The scene, rather than an individual pixel or fastener, is the statistical sampling unit. Shared-input comparisons are paired; the six producer configurations are not six independent samples.

Each input independently draws one of three equally probable geometry families. The surface is $z = (ax^2 + by^2)/2$: $(a, b) = (0, 0)$ for a plane, $(1/R_1, 0)$ for single curvature, and $(1/R_1, 1/R_2)$ for double curvature. Radii are $R_1 \sim \mathcal{U}(2.5, 5)$ m and $R_2 \sim \mathcal{U}(4, 8)$ m. Camera coordinates have $x \sim \mathcal{U}(-0.2, 0.2)$ m, $y \sim \mathcal{U}(-0.1, 0.1)$ m and $z \sim \mathcal{U}(0.7, 1.15)$ m; the optical axis points toward the patch origin. CAD addresses form a 7×5 grid with 35-mm spacing. Fastener heads have 2.5-mm radius, with a narrow bright surrounding rim. Panel seams are located in the same CAD coordinates.

Uniform draws set focal length to 1,200–1,600 pixels, exposure to 3–7 ms, tangential motion to 0.06–0.18 m/s, brightness to 0.85–1.15 and Gaussian intensity-noise standard deviation to 1.5–3.0. The independent light direction, texture phase and camera pose determine the ray/surface image. A diffuse term, specular exponent 36, two brushed-texture sinusoids and CAD-defined seams determine intensity. Horizontal Gaussian exposure filtering uses the median predicted blur divided by $\sqrt{12}$; the vertical scale is 0.18 pixel. The random noise stream is common for paired configurations of the same raster size.

Development used eight pilots and an earlier 384-input, prior-only surface evaluation with distinct seed ranges beginning at 202611100 and 202612100. That evaluation informed the inclusion of pixel-based pose refinement. Its inputs are excluded from the controlled calibration and test sets. The final model, request family and seed specification are preserved by the benchmark hash file.

Table S24: Independent scene counts by geometry family.

Family	Calibration	Test
Planar	68	72
Singly curved	69	59
Doubly curved	55	61

S5.2. Pixel extraction, pose refinement and quality scores

The decoder reads RGB pixels without CAD coordinates. Local contrast is the difference between a Gaussian-smoothed grayscale image and that image. Connected components above contrast 18 and with area 5–240 pixels supply intensity-weighted centers. Predicted CAD positions from the pose prior are subsequently assigned to these centers by minimum-cost bipartite matching with a 30-pixel gate.

The prior translation perturbation is isotropic normal with 3-mm component standard deviation. With probability 0.02, an independent normal perturbation of 5-mm component standard deviation is added. Rotation-vector errors have 0.001-rad component standard deviation. At least six matches trigger nonlinear refinement of six pose parameters. SciPy least-squares applies a componentwise Huber loss with transition 2 to normalized pixel, translation and rotation residuals. Their scales are 0.5 pixel, 3 mm and 0.001 rad. The numerical solver uses at most 60 function evaluations and 10^{-9} stopping tolerances. The protocol's Gaussian-prior descriptor refers to prior sampling; the implemented objective robustifies both observation and prior residuals.

Ground truth is accessed after matching and estimation to label quality. Correct-address coverage uses a half-spacing gate around the physical projected address. Reprojection and mapping quantities take the maximum over those correctly addressed matches within one scene. Mapping casts rays at the true physical image locations through the estimated pose and intersects the CAD surface. An empty correct-match set

has infinite reprojection and mapping quantities; low coverage is independently included in acquisition or association requests. The reported medians and percentiles then summarize these per-scene maxima. They do not summarize all fasteners as independent observations.

The decoder intervention adds 12 pixels to decoded horizontal coordinates before matching and refinement. It represents a changed coordinate convention. It alters the fitted mapping while leaving the rendered observation and optical sampling unchanged. The model retains exact CAD geometry, an ideal perspective camera and prescribed surface appearance; no lens distortion, deformation or physical defect labels enter the numerical scores.

S5.3. Requests and calibration family

The eight requests are compiled through the retained unit-aware request compiler. Millimetres are converted to metres, pixels remain dimensionless image coordinates, and coverage is a ratio. Calibration uses one scene-level union failure for each request. All 48 producer/request events retain allocation $\alpha = 0.05/48$, including events whose cards are not needed by a particular renewal route.

Table S25: Complete request specifications. A dash denotes an omitted quantity.

Request	GSD (mm/px)	Blur (px)	Coverage	Residual (px)	Mapping (mm)
Optical	1.2	3	–	–	–
Optical relaxed	1.5	4	–	–	–
Acquisition	1	2	0.8	–	–
Acquisition relaxed	1.2	3	0.6	–	–
Association	–	–	0.8	8	8
Association relaxed	–	–	0.6	12	12
Complete	1	2	0.8	8	8
Complete relaxed	1.2	3	0.6	12	12

Coverage is a lower bound; the other entries are upper bounds. Complete requests combine acquisition and association at the same strictness. Four strict events constitute the initial anchors. The complete risk-card catalogue follows. Test failures are descriptive held-out outcomes and are not used to construct the bounds.

Table S26: All 48 controlled-study risk cards and held-out outcomes.

Producer	Request	k	n	U	Test failures	Decision
Baseline	Optical	0	192	0.035133	0/192	Admit
Baseline	Optical relaxed	0	192	0.035133	0/192	Admit
Baseline	Acquisition	3	192	0.066020	4/192	Admit
Baseline	Acquisition relaxed	0	192	0.035133	0/192	Admit
Baseline	Association	0	192	0.035133	0/192	Admit
Baseline	Association relaxed	0	192	0.035133	0/192	Admit
Baseline	Complete	3	192	0.066020	4/192	Admit
Baseline	Complete relaxed	0	192	0.035133	0/192	Admit
Focal +15%	Optical	0	192	0.035133	0/192	Admit
Focal +15%	Optical relaxed	0	192	0.035133	0/192	Admit
Focal +15%	Acquisition	5	192	0.082905	8/192	Admit
Focal +15%	Acquisition relaxed	0	192	0.035133	1/192	Admit
Focal +15%	Association	0	192	0.035133	1/192	Admit
Focal +15%	Association relaxed	0	192	0.035133	1/192	Admit
Focal +15%	Complete	5	192	0.082905	8/192	Admit
Focal +15%	Complete relaxed	0	192	0.035133	1/192	Admit

Producer	Request	k	n	U	Test failures	Decision
Exposure +50%	Optical	3	192	0.066020	4/192	Admit
Exposure +50%	Optical relaxed	0	192	0.035133	0/192	Admit
Exposure +50%	Acquisition	22	192	0.201326	34/192	Refuse
Exposure +50%	Acquisition relaxed	3	192	0.066020	4/192	Admit
Exposure +50%	Association	0	192	0.035133	0/192	Admit
Exposure +50%	Association relaxed	0	192	0.035133	0/192	Admit
Exposure +50%	Complete	22	192	0.201326	34/192	Refuse
Exposure +50%	Complete relaxed	3	192	0.066020	4/192	Admit
Wide ROI	Optical	0	192	0.035133	0/192	Admit
Wide ROI	Optical relaxed	0	192	0.035133	0/192	Admit
Wide ROI	Acquisition	3	192	0.066020	4/192	Admit
Wide ROI	Acquisition relaxed	0	192	0.035133	0/192	Admit
Wide ROI	Association	0	192	0.035133	0/192	Admit
Wide ROI	Association relaxed	0	192	0.035133	0/192	Admit
Wide ROI	Complete	3	192	0.066020	4/192	Admit
Wide ROI	Complete relaxed	0	192	0.035133	0/192	Admit
Lower light	Optical	0	192	0.035133	0/192	Admit
Lower light	Optical relaxed	0	192	0.035133	0/192	Admit
Lower light	Acquisition	3	192	0.066020	4/192	Admit
Lower light	Acquisition relaxed	0	192	0.035133	0/192	Admit
Lower light	Association	0	192	0.035133	0/192	Admit
Lower light	Association relaxed	0	192	0.035133	0/192	Admit
Lower light	Complete	3	192	0.066020	4/192	Admit
Lower light	Complete relaxed	0	192	0.035133	0/192	Admit
Decoder shift	Optical	0	192	0.035133	0/192	Admit
Decoder shift	Optical relaxed	0	192	0.035133	0/192	Admit
Decoder shift	Acquisition	3	192	0.066020	4/192	Admit
Decoder shift	Acquisition relaxed	0	192	0.035133	0/192	Admit
Decoder shift	Association	110	192	0.681285	106/192	Refuse
Decoder shift	Association relaxed	0	192	0.035133	0/192	Admit
Decoder shift	Complete	113	192	0.695807	110/192	Refuse
Decoder shift	Complete relaxed	0	192	0.035133	0/192	Admit

S5.4. Complete numerical and computational results

Table S27: Held-out per-scene quality. Each producer has 192 inputs. GSD and blur are medians; coverage is a mean; residual and mapping entries are medians.

Producer	GSD (mm/px)	Blur (px)	Coverage	Residual (px)	Mapping (mm)
Baseline	0.703	0.863	0.99955	0.217	0.056
Focal +15%	0.612	0.993	0.99479	0.203	0.045
Exposure +50%	0.703	1.295	0.99955	0.218	0.056
Wide ROI	0.703	0.863	0.99970	0.219	0.055
Lower light	0.703	0.863	0.99955	0.223	0.056
Decoder shift	0.703	0.863	1.00000	0.261	8.217

Catalogue construction. The three fair methods receive identical stored quantities. Each cache computes an event bound once for an eligible producer/event identity and counts every constructed card, including refusals. Whole-producer renewal constructs 48 cards; dependency-aware exact renewal constructs 42; containment

Table S28: Held-out tail errors and complete-pipeline CPU medians. CPU includes rendering, extraction, estimation and offline scoring; it is distinct from online component timing.

Producer	Residual p95 (px)	Mapping p95 (mm)	CPU median (ms)
Baseline	0.305	0.115	106.702
Focal +15%	0.282	0.102	92.994
Exposure +50%	0.302	0.115	93.198
Wide ROI	0.307	0.116	196.415
Lower light	0.300	0.113	97.168
Decoder shift	0.363	10.673	104.074

with exact fallback constructs 25. All admit the same 44 contexts. Their record evaluations are 9,216, 8,064 and 4,800. A record visit is one scene-level event evaluation, not a new rendering or an individual fastener evaluation. Median process CPU is measured across 11 complete catalogue builds, with records already loaded. Re-execution retains the first timing archive and independently checks decisions and work counts.

The contract constructs strict anchor cards before evaluating requests. If no applicable anchor satisfies the allowance, it builds the exact-event card. The four additional relaxed-event fallbacks occur for acquisition and complete service under exposure change, and association and complete service under decoder change. Cards required for failed strict requests remain in the construction count. The stale-cache control uses baseline cards for all updates and issues four statements unsupported by the fresh catalogue.

Cold source-renewal ledger. The cold workload consists of five updates times eight independently initialized requests. Each case has a 192-input renewal block. It is a logical work ledger evaluated from the registered dependency projection, rather than a measured sum of elapsed times. It does not assume that a batched catalogue rerenders shared quantities separately for every request. The whole-producer method counts 7,680 full pipelines. Both dependency-aware methods count 5,760 full pipelines, 768 optical-only evaluations and 1,152 retained inputs. An executable optical-only implementation independently reproduces GSD and blur for all 1,152 calibration/configuration combinations to tolerance 10^{-10}.

Table S29: Cold source-renewal work aggregated by update. Eight requests and 192 inputs per update.

Update	Whole pipelines	Selected pipelines	Optical only	Retained inputs
Focal +15%	1,536	1,152	384	0
Exposure +50%	1,536	1,152	384	0
Wide ROI	1,536	1,152	0	384
Lower light	1,536	1,152	0	384
Decoder shift	1,536	1,152	0	384

Reproduction and provenance. The directory `studies/cad_surface_20261014` contains the producer, input specification, hash file, sampling and execution driver, complete observations, configurations, risk cards, request plans and work ledgers. Nine compressed image records retain actual rendered pixels and derived centers. Gallery seeds 202616100, 202616104 and 202616108 are the first test input in each family. The full independent rerun compares numerical observations and image records; only machine-dependent CPU fields are excluded from equality. All primary graphics are generated from those saved outputs. The study uses Python 3.12, NumPy, SciPy and Linux on an AMD EPYC 9V74 host, with two workers and one numerical thread per worker.

S6. Extended numerical methods and evidence properties

This section describes the complementary numerical localization and execution-history study. It retains the detailed numerical interface and the conditions supporting the execution and evidence properties. Its synthetic populations and calibration families are separate from the independent CAD-surface benchmark in the main article.

S6.1. Joint-state propagation and checkpoint covariance

The 18-dimensional local perturbation includes robot translation and rotation, six motion biases, and map translation and rotation. Propagation uses

$$x_{k+1} = f(x_k, u_k), \qquad P_{k+1} = F_k P_k F_k^\top + Q_k. \tag{S1}$$

The bias follows a first-order Gauss–Markov model. Robust registration uses a fixed source prior, Huber residuals, an unmatched-point penalty and backtracking fractions $1, 1/2, 1/4, 1/8, 1/16$. The checkpoint covariance is

$$P_s^{(j)} = \mathrm{PSD}\left[(G_j^\top (P_s^-)^{-1} G_j + H_j^\top \Omega_j H_j)^{-1}\right]. \tag{S2}$$

The Jacobians include local rotation perturbations. This covariance is a curvature approximation; empirical risk calibration uses complete executions. A final rejection after an eligible interim publication emits an ordered original-prior rollback. Source replacement begins replay from that source candidate's state and covariance, using only acquired increments. The one-active-source service drains computation and pending publication before admitting another source.

S6.2. Installation work and selection

For publication j, let c_j be total sender work including the common front end, $r_{<j}$ accumulated receiver work, r_0 fixed receiver work and h propagation work per tick. With queue delay $q = 0.05$ s, time step $\Delta t = 0.1$ s and load α, the controlled installation clock uses

$$n_j = \max\left(1, \left\lceil \frac{q + \alpha(c_j + r_{<j} + r_0)}{\Delta t - \alpha h} \right\rceil\right), \qquad r_{<j+1} = r_{<j} + r_0 + n_j h. \tag{S3}$$

The denominator is positive for the frozen services. Next-stage reservation uses the common upper operation count before starting a registration iteration. The sender cap includes numerical and decision work; common front-end, stream and receiver work remain in total availability and work. Mean-characterized operation coefficients define this simulation service. Component process CPU and whole-online-route CPU are reported separately.

The typed record is $\mathcal{K} = (\mathcal{V}, \tau, \mathcal{R}, \sigma, \Gamma)$: entities, entity types, relations, immutable specifications and executable predicates. A before-sensing request $R = (g, a, \mathcal{E}, \delta)$ fixes the population, service, event set and allowance. Selection uses

$$\Pi(R) = \{\pi \in \Pi_0 : \mathrm{match}(g, a, \pi, \nu) \wedge U_{g,a,\pi,e} \le \delta \text{ for all } e \in \mathcal{E}\}, \tag{S4}$$

$$\pi^*(R) = \arg\min_{\pi \in \Pi(R)} \left(\overline{L}_{g,a,\pi} + \lambda_{\mathrm{sel}} \overline{W}_{g,a,\pi}\right). \tag{S5}$$

Policy-ID order resolves ties. An empty set refuses statistical admission. Precommit, actual source sealing, version checking, replay and claim issuance execute on the selected path. The original numerical service remains available with provisional evidence status after detected invalid attachment history.

S6.3. Retained fitted models and policy controls

The retained CART/RF fit uses 926 training and 194 validation sources from a 1,120-source development set. Causal inputs are prior uncertainty, weakest scaled information eigenvalue, residual/inlier measures, step sizes, objective ratio, iteration, pair count, displacement and declared noise. Reachable terminal caps are $\{2, 4, 8, 16, 25\}$. Heuristic predictions estimate terminal MSE, development-event frequency, remaining work and current-versus-prior MSE. Effective terminal labels include rejection and rollback. Publication compares eligible candidate MSE with the prior; selecting an already reached cap from iteration two stops computation. Empty time-feasible sets or profile constraints may also stop the job.

The 22-policy catalogue contains contract-CART, FlatCART, FlatRF and ContractRF at prices 0.3/1/3; Fixed2/4/8/16, Final, Residual, Random2or4 and OldRAVC; and price-one CART/RF time-only interventions. KG denotes the archived contract-CART name. Time-only controls add the same deadline predicate to the flat forecast; the contract adds the declared-profile predicate next. All use the shared stage budget wrapper and acquisition/replay semantics. The time screen predicts delivery using conditional mean remaining work and candidate replay, omitting previously accumulated receiver work; actual event scoring uses the full installation clock. The profile screen caps refinement at four iterations when the declared noise multiplier exceeds two, source position standard deviation exceeds 0.061 m, or geometry was excluded from predictor fitting.

OldRAVC uses the earlier six-output depth-six CART, fitted on 480 training and 120 validation sources. It retains the binary C172 feature. An eligible candidate publishes when forecast event frequency is at most 0.30 and forecast MSE improves on the prior. From iteration two, it stops when predicted four-second MSE gain is no greater than remaining-work cost at price 1 m^2, with the same event-frequency screen. Residual stops at the first eligible checkpoint from iteration two. Random2or4 chooses cap four with frozen continuation probability 0.4654427646, otherwise cap two, using the archived seed-hash coin. Its iid binomial interpretation includes an ideal independent randomization stream. Fixed caps publish an eligible reached cap; Final waits for solver termination. Budget interruption is included in their effective execution maps.

S6.4. Source replacement and delivery exposure

Proposition 1 (Source replacement)**.** *Assume an immutable scan/prior, a retained solve unchanged by publication and no interleaved independent update. Installing the final candidate at a fixed time gives the same live state and covariance whether intermediate candidates were delivered or the final candidate was delivered alone. The same statement holds for ordered prior rollback.*

Proof. The final candidate is the same function of the scan, prior and executed prefix in both schedules. Both receivers assign its replay using the same acquired motion. Intermediate versions contribute no second likelihood factor. □

Proposition 2 (Delivery exposure)**.** *For schedules sharing candidates, exogenous motion and final installation τ_f, let one publish an intermediate candidate at $\tau_c < \tau_f$ and the other hold it. Under integrability,*

$$\mathbb{E}[L_{\text{hold}} - L_{\text{publish}}] = \int_0^T \mathbb{E}\big[\mathbf{1}\{\tau_c \le t < \tau_f\}\left(\|e_0(t)\|^2 - \|e_c(t)\|^2\right)\big]\, dt. \tag{S6}$$

Proof. Estimates coincide before intermediate installation and after the aligned final installation. Subtract the loss integrals pathwise and exchange integration and expectation. □

Eligibility alone does not determine the sign of the exposure difference. Replay has cost. The executed held-publication intervention recomputes delivery timing and may change final installation, so its empirical contrast includes that change.

S6.5. Simultaneous coverage and attachment noninterference

Theorem 3 (Finite-family population-event coverage)**.** *Freeze K complete policy/population/service/event combinations before their calibration outcomes. Assume iid complete inputs from the deployment mixture for each event, including independent policy randomization, and fixed effective execution maps. Exact one-sided binomial bounds with undercoverage allocation β/K cover every risk simultaneously with probability at least $1-\beta$. On that confidence event, calibration-dependent selection among the admitted combinations satisfies its requested risk allowances.*

Proof. Each fixed combination has a binomial failure count and exact inversion undercoverage at most β/K. A union bound establishes simultaneous coverage. Every admitted selection then has its required bounds on that event. Dependence across methods, services and events on a shared input is permitted. □

Confidence probability concerns uncertainty over calibration; the allowance concerns a deployment event. A directly calibrated union differs from two marginal checks. For M episodes the generic mission bound is only $\min(1, M\delta)$ absent further structure. Geometry subgroups and receipt-conditioned subsets require their own statements.

Proposition 4 (Attachment noninterference)**.** *Let $\mathcal{P}$ project a canonical execution $B(X)$ onto numerical decisions, produced/installed versions, replay/rollback order, modeled work, loss, readiness and events. Suppose an observer W calls the original validation/installation with the same arguments and results, changes only separate claim state and excludes its native timing from the frozen modeled clock. Then*

$$\mathcal{P}(W(B(X))) = \mathcal{P}(B(X)). \tag{S7}$$

Canonical valid histories retain original receipts. A detected invalid attempt permanently removes the upper bound in that source scope; a later valid numerical version may still install.

Proof. Induction over validation and installation preserves every original numerical/clock step. Separate claim updates change none of the projected state. The history latch is a persistent conjunction, so a failed attempt cannot reset it within the source scope. □

This projection preserves the canonical calibrated event map. It does not calibrate an arbitrary faulty-history population or a subset selected by remaining receipts. A changed modeled installation schedule requires new calibration.

S6.6. Requirement closure and producer transfer

Proposition 5 (Coverage of a requirement closure)**.** *On the simultaneous confidence event of fixed anchors, an anchor A whose success set is contained in that of R gives $\Pr(E_{p,R}) \le U_{p,A}$. This holds for every requirement in the anchor's pointwise containment closure.*

Proof. Success-set containment gives failure-set containment $E_{p,R} \subseteq E_{p,A}$. Taking probabilities and applying the simultaneous anchor bound proves the statement. Derived requirements add no union-bound allocation. □

Proposition 6 (Invariant requested projection)**.** *Suppose a valid old anchor covers R, the complete-input population is unchanged and every requested function satisfies $z_{p',j}(X) = z_{p,j}(X)$ for all supported inputs. Then the old anchor also bounds $\Pr(E_{p',R})$ on the same confidence event.*

Proof. Requested-function equality yields identical request events, including nonfinite-output failure semantics. Combine that equality with old-producer containment and the anchor bound. □

The premise is functional invariance in the trusted implemented model. Independent A/B pools are used only for such invariant events and retain both source-block identities. Before-sensing commitment is essential: choosing after observing the source between two disjoint 0.10-risk events can produce a selected event with risk 0.20. A directly calibrated switching policy or a common covering event would be needed for that alternative.

S6.7. Unique event support and analytical report timing

The numerical-study audit adds two descriptive checks without changing its labels, cards or decisions. Deduplication uses (study, population, service, event_id). Each 192-label catalogue contains 160 unique event contexts. Contract A/B support is 24/23; direct-renewal support is 30/29; exact-card support is 18/17. Label support remains 24/26, 30/32 and 18/20, respectively.

Report timing satisfies $a_{\text{image}} \geq 0.05 + \alpha w_{\text{camera}}$ and $t_{\text{report}} \geq c + 0.05 + \alpha w_{\text{camera}}$. With frozen camera work of 0.0625 s in A and 0.125 s in B, the image-age lower bounds are A: 0.675/1.3/2.55/3.8 s and B: 1.3/2.55/5.05/7.55 s across relaxed/standard/tight/contended services. Five combinations necessarily violate the full-report 1.8-s image-age limit. Remaining combinations still require acquisition, association and actual timing success.

S7. Earlier localization and image-task studies: sampling and records

This section retains the earlier numerical localization study and two diagnostic image-task studies. Their inputs and calibration families are separate from the sequential campaign and surface challenge in the main article. A source seed is the sampling unit. Repeated policies, services, events and requests are paired executions of that source. Their change/recovery validation is a deterministic software study on existing evidence; it contributes no new population samples or calibration cards.

Table S30: Independent source blocks. Nominal and unfitted populations have disjoint seed ranges. Each stress block contains four explicitly unsupported strata.

Study	Calibration	Confirmation	Stress	Reserved event family
Localization	8,192	4,096	1,536	528
Image A, 192×144	4,096	2,048	256	4,608 jointly
Image B, 288×216	4,096	2,048	256	

The localization source blocks start at 8,300,000/8,400,000 for calibration and 8,500,000/8,600,000 for confirmation. Its four 384-source stress strata start at 8,700,000, 8,800,000, 8,900,000 and 9,000,000. The 48 excluded pilot sources start at 8,200,000. The retained predictor fit has 926 training and 194 validation sources.

For image A, calibration starts at 13,000,000/13,100,000 and confirmation at 13,200,000/13,300,000. Image B uses 14,000,000/14,100,000 and 14,200,000/14,300,000, respectively. Each image population contains 2,048 calibration and 1,024 confirmation sources. Stress starts at 13,400,000 or 14,400,000, with an additional 1,000 per successive stratum and 64 inputs per stratum. The 48 excluded image pilots start at 12,000,000. There are 409,600 paired policy/service outcomes and 6,291,456 repeated confirmation requests across A/B; these counts do not increase the independent source count.

Nominal draws mix B767/C172, position-prior deviations 0.015/0.03/0.06 m, range-noise multipliers 1/2 and shape levels 0/1/2. Source frames are sampled from prescribed trajectories, excluding the first and last 100 frames. Map translation and rotation perturbations have standard deviations 0.025 m and 0.001 rad. The unfitted procedural airframe is absent from predictor training and has independently matched calibration. Stress separately removes a 2.2-rad range sector, retains a 0.5-rad sector, increases prior deviation to 0.08/0.12 m, or introduces a 0.06-m map warp.

Numerical methods, generators, fitted predictors, event definitions and characterized clocks were sealed at F0 before formal generation. Calibration cards and before-sensing plans were sealed at F1 before confirmation. Versioned amendments are preserved: localization source-history monitoring changed only attachment state, with audited numerical/clock/event noninterference; image manifest formatting preserved scientific parameters; conservative attachment dependencies added capture/motion/geometry providers; direct-renewal parity restored access to applicable old evidence. The final image family remains 4,608 and unused allocation remains reserved. The archived hashes establish local execution chronology.

Two missing localization index records were recovered on their original seeds with identical numerical arrays; their recovered CPU fields are new timing measurements. Continuous routes contain 36 distinct 600-s

inputs, with four matched realizations for each geometry/condition and 12 methods, yielding 432 executions. The online timer excludes generation, archive hydration and truth scoring. The retained original manuscripts, superseded protocols and earlier A19/A20/R4/R5 results are separately archived; they do not enter the sample counts or tables for the localization and diagnostic image studies presented here.

S8. Image requirements, dependencies and timing

The acquisition event includes fixed-target GSD, exposure blur and acquired target coverage. Association includes coverage, reprojection, metric displacement and a positive version at capture. Reporting combines both tasks with delay and image age. Their limits define synthetic diagnostic services; palette landmarks evaluate acquisition and association within the implemented image producer.

Table S31: Canonical full, partial and strict thresholds. Blank bounds denote a quantity omitted from that service. Reprojection and metric association errors retain different units.

Quantity	Full	Partial	Strict
GSD (mm/pixel)	20	40	10
Blur (pixel)	0.50	0.75	0.35
Coverage (fraction)	0.50	0.20	0.70
Reprojection (pixel)	30	50	15
Association (m)	0.10	0.15	0.06
Report delay (s)	2	3	1
Image age (s)	1.8	2.8	0.8

Partial-relaxed removes GSD and reprojection bounds and uses delay/age 4/3.8 s. Coarser-GSD, more-blur and later-report variants change only the named full threshold to 40 mm/pixel, 0.75 pixel or 4/3.8 s. Acquisition's component variant retains GSD 40 and blur 0.75; association's retains displacement 0.15 m and readiness; reporting's retains delay/age 3/2.8 s. These eight named variants contain duplicates for tasks unaffected by a particular edit. There are 192 labels but 160 unique population/service/event contexts per study at each allowance.

The compiler supports length and time conversions, pixel units and ratios. It rejects incompatible dimensions, Boolean/nonfinite thresholds, duplicate quantity intervals and empty intervals. Nonfinite required outputs fail an event. An absent required output raises a schema error and prevents valid evaluation. Thus missing schema fields are distinguished from a defined observation's failure. Preflight checks 10,000 event pairs on one million generated metric vectors and 144 task-specific fixtures; those checks are excluded software cases.

Every requested quantity includes the conservative capture/motion/geometry-provider dependencies from the sealed attachment amendment. The other dependencies follow the executed data flow: GSD includes target plan and focal length; blur additionally includes exposure; coverage includes ROI, rendering, pixel noise and decoding; reprojection/association additionally include localization; readiness includes localization; report delay/age include localization, virtual camera clock and queue. Transfer checks the union of requested dependencies, retaining old calibration lineage. Changes in quantities outside that requested projection do not enter its capsule.

Both ROIs use focal length 1,800 pixels, exposure 6 ms and the same preplanned target definition. The excluded-pilot virtual camera/decoder clocks are 0.0625 s for A and 0.125 s for B. With load 10/20/40/60, image-age lower bounds are A: 0.675/1.3/2.55/3.8 s and B: 1.3/2.55/5.05/7.55 s. Five study/service combinations necessarily violate the full 1.8-s limit. Remaining combinations still require all other task conditions.

The image receipt observer operates on archived numerical records and is timed separately. Its source seal, study, population, policy, event, plan and card checks preserve source-scoped refusal across reconstructed gates. Original fixtures include 72 provider edits and 32 malformed packets. The original source replacement/replay checks execute inside the numerical localization service. Component CPU, whole-route CPU, virtual operation

work and preloaded-record query wall time retain separate boundaries. Windows camera CPU has 15.625-ms quantization.

S9. Inference, sensitivity and data interfaces

Exact binomial inversion uses β/K, with $\beta = 0.05$, $K = 528$ for localization and $K = 4,608$ jointly for A/B. A valid statistical statement assumes iid complete inputs, a fixed executed event map and matched deployment distribution. Correlation across policies, events or services sharing a source is permitted by the simultaneous union-bound argument. Pooling A/B is restricted to explicitly invariant requested projections and preserves both independent manifest identities.

The 16 primary localization contrasts use 2,048 confirmation sources per population and 32,000 paired bootstrap resamples. Pointwise and family-adjusted percentile intervals are approximate sampling intervals. They are distinct from exact binomial admission bounds. Nested prefixes, alternative confidence levels, geometry subgroups, Pareto fronts, stress and complete routes are descriptive; they do not revise F1 or obtain additional conditional guarantees.

The accompanying CSV files retain full-precision cards, all 6,144 plans, confirmation summaries, all geometry/service/policy means, unsupported stress, route results and software case decisions. The following tables document the questions evaluated in these earlier studies. Complete population source generation and raw arrays remain in the original verified release assets. The manuscript package includes sealed evidence, five complete archived replay inputs, table snapshots, deterministic figure scripts, the typed compiler, producer-audit and case-validation code, and SHA-256 manifests. Submitted figures use original photographs, executed synthetic rasters and deterministic vector graphics.

S10. Executed change and recovery validation

The supplementary software study uses all 320 unique primary A/B request contexts. The original semantic policy is held fixed; a direct-renewal policy supplies the fixed alternative when that semantic request was refused, otherwise Fixed2 is the prespecified control. Sixteen configurations per context comprise no change, each of 13 single provider changes, and ROI/clock and decoder/clock pairs. For each method, the validator checks the edited context and restoration of the original trusted producer. An independently written oracle inspects intervals and the same amended dependency specification without calling the hash or containment routines used by the implementations.

Three complete methods receive identical applicable card inventories and cached evidence: contract coverage, independently coded procedural coverage and cached direct renewal. Stale-plan reuse and whole-producer invalidation are separately identified diagnostic controls. Incorrect retention and unnecessary withdrawal are defined relative to the same method's admissible archived evidence. These are deterministic conformance counts, with zero new geometric draws and zero new calibration cards. A restored producer uses its original specification and a fresh trusted execution scope after a detected history fault.

The concrete source is the previously archived replay seed 14,300,000, selected in the earlier reproduction handoff. Its RGB array, numerical state arrays and execution metadata are hash checked. FlatCART at relaxed service installs iterations one and two at 0.1/0.2 s, and capture at 0.2 s uses the second version. The full-association anchor has $k/n = 229/2,048$ and $U = 0.1439731071$. A predeclared reprojection relaxation from 30 to 50 pixels lies in its containment closure. A decoder edit withdraws full association while preserving a covered readiness-only requirement. The optical component remains refused under the unfitted-population risk bound. Restoring the original decoder restores the full-association plan; packet history faults remain refused until a fresh trusted execution scope.

The first illustrative assertion expected the optical component to be admitted. Its actual archived bound rejected that expectation. The failed source/freeze and refinement note are retained, the optical refusal stays visible, and the readiness-only derived requirement was added to demonstrate existing component evidence under the unchanged requested projection. The 320-by-16 change matrix, numerical observations and risk allocations were unchanged. Query timing covers preloaded-card revalidation and restoration only.

Matched-support record counts measure the effort to construct the required proof catalogues; warmed caches in all complete methods require zero further record rescoring.

S10.1. Conformance and common-support accounting

Table S32: Executed deterministic software decisions over the same prespecified change contexts. Counts are conformance fixtures, not independent population samples or fault-rate estimates.

Method	Decisions	Missed	Unnecessary	Restore
Contract	5120	0	0	0
Cached direct	5120	0	0	0
Complete procedure	5120	0	0	0

Table S33: Matched support with the same precommitted policies. Counted work to construct selected proof catalogues from archived calibration records; it excludes source acquisition and producer execution. Both complete implementations share anchor accounting.

Study	Method	Requests	Cards	Record visits
A	Anchor reuse	24	12	24576
A	Direct events	24	15	30720
B	Anchor reuse	23	11	24576
B	Direct events	23	14	30720
combined	Anchor reuse	47	23	49152
combined	Direct events	47	29	61440

S11. Complete localization confirmation catalogue

Table S34: All 22 policies, both populations and all four services. Loss L is 10^{-3} m^{2}s; modeled work W is ms; failure columns are percentages. Every source is retained. Native CPU and full tails are in the accompanying CSV.

Population	Service	Policy	n	L	W	Obs.	Union
unfitted	contended	CART_Time	2048	22.353	7.616	77.25	77.98
unfitted	contended	ContractRF	2048	22.405	7.869	92.43	92.77
unfitted	contended	ContractRF0.3	2048	22.405	7.869	92.43	92.77
unfitted	contended	ContractRF3	2048	22.405	7.869	92.43	92.77
unfitted	contended	Final	2048	18.559	12.001	100.00	100.00
unfitted	contended	Fixed16	2048	18.559	12.001	100.00	100.00
unfitted	contended	Fixed2	2048	20.763	9.453	100.00	100.00
unfitted	contended	Fixed4	2048	18.559	12.001	100.00	100.00
unfitted	contended	Fixed8	2048	18.559	12.001	100.00	100.00
unfitted	contended	FlatCART	2048	18.173	11.199	79.15	79.69
unfitted	contended	FlatCART0.3	2048	17.967	11.653	79.15	79.64
unfitted	contended	FlatCART3	2048	19.078	10.148	79.15	79.64
unfitted	contended	FlatRF	2048	20.523	10.344	92.72	92.92
unfitted	contended	FlatRF0.3	2048	20.523	10.344	92.72	92.92
unfitted	contended	FlatRF3	2048	20.523	10.344	92.72	92.92

Population	Service	Policy	n	L	W	Obs.	Union
unfitted	contended	KG0.3	2048	22.353	7.616	77.25	77.98
unfitted	contended	KG1	2048	22.353	7.616	77.25	77.98
unfitted	contended	KG3	2048	22.353	7.616	77.25	77.98
unfitted	contended	OldRAVC	2048	18.611	10.139	82.91	83.06
unfitted	contended	RF_Time	2048	22.405	7.869	92.43	92.77
unfitted	contended	Random2or4	2048	19.659	10.623	100.00	100.00
unfitted	contended	Residual	2048	20.417	9.853	76.81	77.44
unfitted	relaxed	CART_Time	2048	17.095	10.598	8.01	8.79
unfitted	relaxed	ContractRF	2048	17.086	11.054	8.35	8.94
unfitted	relaxed	ContractRF0.3	2048	16.671	11.917	8.25	8.94
unfitted	relaxed	ContractRF3	2048	18.503	10.063	8.89	9.62
unfitted	relaxed	Final	2048	17.030	11.688	8.74	9.47
unfitted	relaxed	Fixed16	2048	17.030	11.688	8.74	9.47
unfitted	relaxed	Fixed2	2048	19.937	9.219	9.13	10.06
unfitted	relaxed	Fixed4	2048	17.030	11.688	8.74	9.47
unfitted	relaxed	Fixed8	2048	17.030	11.688	8.74	9.47
unfitted	relaxed	FlatCART	2048	17.095	10.598	8.01	8.79
unfitted	relaxed	FlatCART0.3	2048	16.859	10.998	8.20	8.89
unfitted	relaxed	FlatCART3	2048	18.129	9.667	8.54	9.23
unfitted	relaxed	FlatRF	2048	17.086	11.054	8.35	8.94
unfitted	relaxed	FlatRF0.3	2048	16.671	11.917	8.25	8.94
unfitted	relaxed	FlatRF3	2048	18.503	10.063	8.89	9.62
unfitted	relaxed	KG0.3	2048	16.859	10.998	8.20	8.89
unfitted	relaxed	KG1	2048	17.095	10.598	8.01	8.79
unfitted	relaxed	KG3	2048	18.129	9.667	8.54	9.23
unfitted	relaxed	OldRAVC	2048	17.717	9.718	8.35	9.03
unfitted	relaxed	RF_Time	2048	17.086	11.054	8.35	8.94
unfitted	relaxed	Random2or4	2048	18.535	10.348	9.13	10.06
unfitted	relaxed	Residual	2048	19.609	9.388	8.98	9.91
unfitted	standard	CART_Time	2048	17.423	10.702	7.91	8.79
unfitted	standard	ContractRF	2048	19.936	9.953	9.08	10.06
unfitted	standard	ContractRF0.3	2048	19.936	9.953	9.08	10.06
unfitted	standard	ContractRF3	2048	19.936	9.953	9.08	10.06
unfitted	standard	Final	2048	17.350	11.736	8.54	9.28
unfitted	standard	Fixed16	2048	17.350	11.736	8.54	9.28
unfitted	standard	Fixed2	2048	19.938	9.220	9.08	10.06
unfitted	standard	Fixed4	2048	17.350	11.736	8.54	9.28
unfitted	standard	Fixed8	2048	17.350	11.736	8.54	9.28
unfitted	standard	FlatCART	2048	17.423	10.702	7.91	8.79
unfitted	standard	FlatCART0.3	2048	17.194	11.113	8.15	8.89
unfitted	standard	FlatCART3	2048	18.431	9.750	8.54	9.23
unfitted	standard	FlatRF	2048	19.936	9.953	9.08	10.06
unfitted	standard	FlatRF0.3	2048	19.936	9.953	9.08	10.06
unfitted	standard	FlatRF3	2048	19.936	9.953	9.08	10.06
unfitted	standard	KG0.3	2048	17.194	11.113	8.15	8.89
unfitted	standard	KG1	2048	17.423	10.702	7.91	8.79
unfitted	standard	KG3	2048	18.431	9.750	8.54	9.23
unfitted	standard	OldRAVC	2048	17.964	9.781	8.40	9.03
unfitted	standard	RF_Time	2048	19.936	9.953	9.08	10.06
unfitted	standard	Random2or4	2048	18.675	10.375	9.03	9.96
unfitted	standard	Residual	2048	19.857	9.469	8.94	9.91
unfitted	tight	CART_Time	2048	22.172	7.552	94.87	95.12
unfitted	tight	ContractRF	2048	22.195	7.795	99.95	99.95
unfitted	tight	ContractRF0.3	2048	22.195	7.795	99.95	99.95
unfitted	tight	ContractRF3	2048	22.195	7.795	99.95	99.95

Population	Service	Policy	n	L	W	Obs.	Union
unfitted	tight	Final	2048	22.168	7.542	94.53	94.78
unfitted	tight	Fixed16	2048	22.168	7.542	94.53	94.78
unfitted	tight	Fixed2	2048	22.168	7.542	94.53	94.78
unfitted	tight	Fixed4	2048	22.168	7.542	94.53	94.78
unfitted	tight	Fixed8	2048	22.168	7.542	94.53	94.78
unfitted	tight	FlatCART	2048	22.172	7.552	94.87	95.12
unfitted	tight	FlatCART0.3	2048	22.172	7.552	94.87	95.12
unfitted	tight	FlatCART3	2048	22.172	7.552	94.87	95.12
unfitted	tight	FlatRF	2048	22.195	7.795	99.95	99.95
unfitted	tight	FlatRF0.3	2048	22.195	7.795	99.95	99.95
unfitted	tight	FlatRF3	2048	22.195	7.795	99.95	99.95
unfitted	tight	KG0.3	2048	22.172	7.552	94.87	95.12
unfitted	tight	KG1	2048	22.172	7.552	94.87	95.12
unfitted	tight	KG3	2048	22.172	7.552	94.87	95.12
unfitted	tight	OldRAVC	2048	22.171	7.549	94.78	95.02
unfitted	tight	RF_Time	2048	22.195	7.795	99.95	99.95
unfitted	tight	Random2or4	2048	22.168	7.542	94.53	94.78
unfitted	tight	Residual	2048	22.168	7.542	94.53	94.78
nominal	contended	CART_Time	2048	17.620	7.583	49.22	49.95
nominal	contended	ContractRF	2048	17.679	7.839	66.16	66.60
nominal	contended	ContractRF0.3	2048	17.679	7.839	66.16	66.60
nominal	contended	ContractRF3	2048	17.679	7.839	66.16	66.60
nominal	contended	Final	2048	12.823	12.377	100.00	100.00
nominal	contended	Fixed16	2048	12.823	12.377	100.00	100.00
nominal	contended	Fixed2	2048	14.802	9.432	100.00	100.00
nominal	contended	Fixed4	2048	12.823	12.377	100.00	100.00
nominal	contended	Fixed8	2048	12.823	12.377	100.00	100.00
nominal	contended	FlatCART	2048	12.527	11.183	50.88	52.39
nominal	contended	FlatCART0.3	2048	12.170	12.053	50.88	52.49
nominal	contended	FlatCART3	2048	13.098	10.098	50.88	52.05
nominal	contended	FlatRF	2048	14.577	10.351	66.85	67.82
nominal	contended	FlatRF0.3	2048	14.577	10.351	66.85	67.82
nominal	contended	FlatRF3	2048	14.577	10.351	66.85	67.82
nominal	contended	KG0.3	2048	17.620	7.583	49.22	49.95
nominal	contended	KG1	2048	17.620	7.583	49.22	49.95
nominal	contended	KG3	2048	17.620	7.583	49.22	49.95
nominal	contended	OldRAVC	2048	13.106	10.007	56.54	57.03
nominal	contended	RF_Time	2048	17.679	7.839	66.16	66.60
nominal	contended	Random2or4	2048	13.608	10.756	100.00	100.00
nominal	contended	Residual	2048	14.481	9.838	48.78	50.00
nominal	relaxed	CART_Time	2048	11.408	10.619	5.76	6.93
nominal	relaxed	ContractRF	2048	11.125	10.990	5.66	6.64
nominal	relaxed	ContractRF0.3	2048	10.706	12.446	6.05	7.13
nominal	relaxed	ContractRF3	2048	12.152	10.031	5.81	7.23
nominal	relaxed	Final	2048	10.911	12.072	6.79	7.81
nominal	relaxed	Fixed16	2048	10.911	12.072	6.79	7.81
nominal	relaxed	Fixed2	2048	13.830	9.221	7.08	8.59
nominal	relaxed	Fixed4	2048	10.911	12.072	6.79	7.81
nominal	relaxed	Fixed8	2048	10.911	12.072	6.79	7.81
nominal	relaxed	FlatCART	2048	11.408	10.619	5.76	6.93
nominal	relaxed	FlatCART0.3	2048	11.002	11.389	5.86	6.88
nominal	relaxed	FlatCART3	2048	12.031	9.646	5.81	7.08
nominal	relaxed	FlatRF	2048	11.125	10.990	5.66	6.64
nominal	relaxed	FlatRF0.3	2048	10.706	12.446	6.05	7.13
nominal	relaxed	FlatRF3	2048	12.152	10.031	5.81	7.23

Population	Service	Policy	n	L	W	Obs.	Union
nominal	relaxed	KG0.3	2048	11.002	11.389	5.86	6.88
nominal	relaxed	KG1	2048	11.408	10.619	5.76	6.93
nominal	relaxed	KG3	2048	12.031	9.646	5.81	7.08
nominal	relaxed	OldRAVC	2048	11.950	9.600	5.42	6.74
nominal	relaxed	RF_Time	2048	11.125	10.990	5.66	6.64
nominal	relaxed	Random2or4	2048	12.254	10.507	7.42	8.50
nominal	relaxed	Residual	2048	13.590	9.403	6.88	8.45
nominal	standard	CART_Time	2048	11.664	10.705	5.52	6.84
nominal	standard	ContractRF	2048	13.844	9.973	7.18	8.64
nominal	standard	ContractRF0.3	2048	13.844	9.973	7.18	8.64
nominal	standard	ContractRF3	2048	13.844	9.973	7.18	8.64
nominal	standard	Final	2048	11.295	12.109	6.49	7.52
nominal	standard	Fixed16	2048	11.295	12.109	6.49	7.52
nominal	standard	Fixed2	2048	13.833	9.221	7.18	8.64
nominal	standard	Fixed4	2048	11.295	12.109	6.49	7.52
nominal	standard	Fixed8	2048	11.295	12.109	6.49	7.52
nominal	standard	FlatCART	2048	11.664	10.705	5.52	6.84
nominal	standard	FlatCART0.3	2048	11.269	11.495	5.62	6.79
nominal	standard	FlatCART3	2048	12.301	9.721	5.76	7.13
nominal	standard	FlatRF	2048	13.844	9.973	7.18	8.64
nominal	standard	FlatRF0.3	2048	13.844	9.973	7.18	8.64
nominal	standard	FlatRF3	2048	13.844	9.973	7.18	8.64
nominal	standard	KG0.3	2048	11.269	11.495	5.62	6.79
nominal	standard	KG1	2048	11.664	10.705	5.52	6.84
nominal	standard	KG3	2048	12.301	9.721	5.76	7.13
nominal	standard	OldRAVC	2048	12.199	9.666	5.42	6.79
nominal	standard	RF_Time	2048	13.844	9.973	7.18	8.64
nominal	standard	Random2or4	2048	12.379	10.520	7.13	8.25
nominal	standard	Residual	2048	13.817	9.476	6.98	8.50
nominal	tight	CART_Time	2048	17.401	7.522	73.44	73.63
nominal	tight	ContractRF	2048	17.474	7.784	96.44	96.44
nominal	tight	ContractRF0.3	2048	17.474	7.784	96.44	96.44
nominal	tight	ContractRF3	2048	17.474	7.784	96.44	96.44
nominal	tight	Final	2048	17.391	7.511	71.73	71.92
nominal	tight	Fixed16	2048	17.391	7.511	71.73	71.92
nominal	tight	Fixed2	2048	17.391	7.511	71.73	71.92
nominal	tight	Fixed4	2048	17.391	7.511	71.73	71.92
nominal	tight	Fixed8	2048	17.391	7.511	71.73	71.92
nominal	tight	FlatCART	2048	17.401	7.522	73.44	73.63
nominal	tight	FlatCART0.3	2048	17.401	7.522	73.44	73.63
nominal	tight	FlatCART3	2048	17.401	7.522	73.44	73.63
nominal	tight	FlatRF	2048	17.474	7.784	96.44	96.44
nominal	tight	FlatRF0.3	2048	17.474	7.784	96.44	96.44
nominal	tight	FlatRF3	2048	17.474	7.784	96.44	96.44
nominal	tight	KG0.3	2048	17.401	7.522	73.44	73.63
nominal	tight	KG1	2048	17.401	7.522	73.44	73.63
nominal	tight	KG3	2048	17.401	7.522	73.44	73.63
nominal	tight	OldRAVC	2048	17.397	7.519	72.85	73.05
nominal	tight	RF_Time	2048	17.474	7.784	96.44	96.44
nominal	tight	Random2or4	2048	17.391	7.511	71.73	71.92
nominal	tight	Residual	2048	17.391	7.511	71.73	71.92

S12. Complete continuous-route catalogue

Table S35: All 12 methods and nine geometry/condition combinations. Each mean summarizes four matched 600-s inputs. CPU covers the complete online loop; route evidence remains descriptive.

Geometry	Condition	Method	RMSE (cm)	Success (%)	CPU (s)
B767	contention	CART_Time	3.543	54.43	2.038
B767	contention	ContractRF	3.523	44.57	2.345
B767	contention	Final	3.605	45.48	2.566
B767	contention	Fixed2	3.555	46.99	2.322
B767	contention	Fixed4	3.605	45.48	2.607
B767	contention	FlatCART	3.581	54.10	2.505
B767	contention	FlatRF	3.549	44.57	2.576
B767	contention	Full	3.531	55.02	2.019
B767	contention	KG1	3.543	54.43	2.246
B767	contention	KG1_Held	3.545	54.43	2.325
B767	contention	OldRAVC	3.550	54.85	2.299
B767	contention	TaskAwareFlatValidated	3.531	55.02	2.158
B767	stress	CART_Time	8.200	55.85	2.213
B767	stress	ContractRF	8.186	47.74	2.136
B767	stress	Final	8.215	39.97	2.668
B767	stress	Fixed2	8.185	42.22	2.154
B767	stress	Fixed4	8.215	39.97	2.557
B767	stress	FlatCART	8.215	53.85	2.583
B767	stress	FlatRF	8.187	47.41	2.450
B767	stress	Full	8.158	56.19	2.300
B767	stress	KG1	8.200	55.85	2.207
B767	stress	KG1_Held	8.205	55.60	2.199
B767	stress	OldRAVC	8.212	53.18	2.440
B767	stress	TaskAwareFlatValidated	8.158	56.19	2.175
B767	ordinary	CART_Time	3.706	96.49	2.767
B767	ordinary	ContractRF	3.805	97.07	2.728
B767	ordinary	Final	3.719	94.57	3.228
B767	ordinary	Fixed2	3.806	97.07	2.450
B767	ordinary	Fixed4	3.719	94.57	3.176
B767	ordinary	FlatCART	3.706	96.49	2.711
B767	ordinary	FlatRF	3.805	97.07	2.596
B767	ordinary	Full	3.715	97.24	2.514
B767	ordinary	KG1	3.706	96.49	2.809
B767	ordinary	KG1_Held	3.712	96.32	2.562
B767	ordinary	OldRAVC	3.722	97.07	2.500
B767	ordinary	TaskAwareFlatValidated	3.715	97.24	2.504
C172	contention	CART_Time	5.341	80.94	2.162
C172	contention	ContractRF	5.256	53.93	2.090
C172	contention	Final	5.388	51.76	2.734
C172	contention	Fixed2	5.237	53.01	2.121
C172	contention	Fixed4	5.388	51.76	2.659
C172	contention	FlatCART	5.321	81.02	2.374
C172	contention	FlatRF	5.236	53.68	2.288
C172	contention	Full	5.234	81.61	2.028
C172	contention	KG1	5.341	80.94	2.194
C172	contention	KG1_Held	5.342	80.85	2.096
C172	contention	OldRAVC	5.193	81.77	2.173
C172	contention	TaskAwareFlatValidated	5.234	81.61	2.230
C172	stress	CART_Time	14.439	11.45	2.054
C172	stress	ContractRF	14.398	6.27	2.342
C172	stress	Final	14.333	10.03	2.754

Geometry	Condition	Method	RMSE (cm)	Success (%)	CPU (s)
C172	stress	Fixed2	14.488	10.20	2.168
C172	stress	Fixed4	14.333	10.03	2.545
C172	stress	FlatCART	14.494	11.62	2.432
C172	stress	FlatRF	14.490	6.19	2.285
C172	stress	Full	14.414	11.20	2.221
C172	stress	KG1	14.439	11.45	2.198
C172	stress	KG1_Held	14.440	11.45	2.120
C172	stress	OldRAVC	14.403	11.45	2.480
C172	stress	TaskAwareFlatValidated	14.414	11.20	2.189
C172	ordinary	CART_Time	4.832	93.06	2.607
C172	ordinary	ContractRF	4.787	94.15	2.675
C172	ordinary	Final	4.836	91.64	2.941
C172	ordinary	Fixed2	4.787	94.15	2.297
C172	ordinary	Fixed4	4.836	91.64	3.189
C172	ordinary	FlatCART	4.832	93.06	2.378
C172	ordinary	FlatRF	4.787	94.15	2.651
C172	ordinary	Full	4.805	93.81	2.518
C172	ordinary	KG1	4.832	93.06	2.589
C172	ordinary	KG1_Held	4.832	93.06	2.562
C172	ordinary	OldRAVC	4.837	93.65	2.267
C172	ordinary	TaskAwareFlatValidated	4.805	93.81	2.491
Unfitted	contention	CART_Time	2.993	43.65	2.027
Unfitted	contention	ContractRF	2.993	40.38	2.319
Unfitted	contention	Final	3.071	39.21	2.694
Unfitted	contention	Fixed2	3.064	39.21	2.108
Unfitted	contention	Fixed4	3.071	39.21	2.353
Unfitted	contention	FlatCART	3.054	43.73	2.382
Unfitted	contention	FlatRF	3.057	40.22	2.322
Unfitted	contention	Full	2.994	43.65	1.975
Unfitted	contention	KG1	2.993	43.65	2.078
Unfitted	contention	KG1_Held	2.996	43.65	1.934
Unfitted	contention	OldRAVC	3.048	43.81	2.383
Unfitted	contention	TaskAwareFlatValidated	2.994	43.65	2.090
Unfitted	stress	CART_Time	9.649	46.49	2.207
Unfitted	stress	ContractRF	9.633	42.56	1.907
Unfitted	stress	Final	9.283	37.04	2.531
Unfitted	stress	Fixed2	9.665	38.21	2.017
Unfitted	stress	Fixed4	9.283	37.04	2.519
Unfitted	stress	FlatCART	9.672	45.07	2.500
Unfitted	stress	FlatRF	9.661	42.14	2.451
Unfitted	stress	Full	9.663	45.74	2.211
Unfitted	stress	KG1	9.649	46.49	2.147
Unfitted	stress	KG1_Held	9.654	46.49	2.042
Unfitted	stress	OldRAVC	9.269	44.82	2.464
Unfitted	stress	TaskAwareFlatValidated	9.663	45.74	2.401
Unfitted	ordinary	CART_Time	3.018	99.25	2.507
Unfitted	ordinary	ContractRF	2.997	99.41	2.673
Unfitted	ordinary	Final	3.022	98.91	2.674
Unfitted	ordinary	Fixed2	2.997	99.41	2.413
Unfitted	ordinary	Fixed4	3.022	98.91	2.688
Unfitted	ordinary	FlatCART	3.018	99.25	2.635
Unfitted	ordinary	FlatRF	2.997	99.41	2.521
Unfitted	ordinary	Full	3.006	99.25	2.395
Unfitted	ordinary	KG1	3.018	99.25	2.618
Unfitted	ordinary	KG1_Held	3.027	99.16	2.577

Geometry	Condition	Method	RMSE (cm)	Success (%)	CPU (s)
Unfitted	ordinary	OldRAVC	3.006	99.25	2.290
Unfitted	ordinary	TaskAwareFlatValidated	3.006	99.25	2.235

S13. Image-task numerical confirmation and stress

Table S36: All eight numerical policies by geometry for the standard service. Total work includes the virtual camera. The complete 192-row all-service table is supplied as CSV; geometry summaries carry no conditional risk certificate.

Study	Geometry	Policy	n	RMSE (cm)	Work (ms)	Union (%)
A	B767	ContractRF	515	4.872	72.816	9.71
A	B767	Final	515	4.147	74.974	7.77
A	B767	Fixed2	515	4.872	72.074	9.71
A	B767	Fixed4	515	4.147	74.974	7.77
A	B767	FlatCART	515	4.327	73.629	6.60
A	B767	FlatRF	515	4.872	72.816	9.71
A	B767	KG1	515	4.327	73.629	6.60
A	B767	OldRAVC	515	4.624	72.595	7.77
A	C172	ContractRF	509	4.609	72.087	6.48
A	C172	Final	509	4.378	74.237	6.68
A	C172	Fixed2	509	4.609	71.330	6.48
A	C172	Fixed4	509	4.378	74.237	6.68
A	C172	FlatCART	509	4.503	72.568	6.09
A	C172	FlatRF	509	4.609	72.087	6.48
A	C172	KG1	509	4.503	72.568	6.09
A	C172	OldRAVC	509	4.571	71.654	6.09
A	Unfitted	ContractRF	1024	4.864	72.454	10.45
A	Unfitted	Final	1024	4.499	74.263	9.47
A	Unfitted	Fixed2	1024	4.864	71.719	10.45
A	Unfitted	Fixed4	1024	4.499	74.263	9.47
A	Unfitted	FlatCART	1024	4.640	73.235	9.57
A	Unfitted	FlatRF	1024	4.864	72.454	10.45
A	Unfitted	KG1	1024	4.640	73.235	9.57
A	Unfitted	OldRAVC	1024	4.728	72.249	9.28
B	B767	ContractRF	513	5.164	135.336	12.48
B	B767	Final	513	4.333	137.553	9.36
B	B767	Fixed2	513	5.164	134.596	12.48
B	B767	Fixed4	513	4.333	137.553	9.36
B	B767	FlatCART	513	4.495	136.455	8.38
B	B767	FlatRF	513	5.164	135.336	12.48
B	B767	KG1	513	4.495	136.455	8.38
B	B767	OldRAVC	513	4.705	135.184	8.58
B	C172	ContractRF	511	4.346	134.588	3.33
B	C172	Final	511	4.154	136.753	4.70
B	C172	Fixed2	511	4.346	133.831	3.33
B	C172	Fixed4	511	4.154	136.753	4.70
B	C172	FlatCART	511	4.326	135.248	3.72
B	C172	FlatRF	511	4.346	134.588	3.33
B	C172	KG1	511	4.326	135.248	3.72
B	C172	OldRAVC	511	4.315	134.111	2.94
B	Unfitted	ContractRF	1024	4.829	134.947	10.55

Study	Geometry	Policy	n	RMSE (cm)	Work (ms)	Union (%)
B	Unfitted	Final	1024	4.427	136.759	8.59
B	Unfitted	Fixed2	1024	4.829	134.212	10.55
B	Unfitted	Fixed4	1024	4.427	136.759	8.59
B	Unfitted	FlatCART	1024	4.552	135.694	8.79
B	Unfitted	FlatRF	1024	4.829	134.947	10.55
B	Unfitted	KG1	1024	4.552	135.694	8.79
B	Unfitted	OldRAVC	1024	4.663	134.805	9.38

Table S37: All standard-service unsupported image stress strata and policies. The full 512-row all-service table is retained as CSV. These observations carry no transferred population-risk statement.

Study	Stress	Geometry	Policy	n	RMSE (cm)	Union (%)
A	degeneracy	B767	ContractRF	28	5.92	14.3
A	degeneracy	B767	Final	28	5.23	17.9
A	degeneracy	B767	Fixed2	28	5.92	14.3
A	degeneracy	B767	Fixed4	28	5.23	17.9
A	degeneracy	B767	FlatCART	28	5.22	17.9
A	degeneracy	B767	FlatRF	28	5.92	14.3
A	degeneracy	B767	KG1	28	5.22	17.9
A	degeneracy	B767	OldRAVC	28	5.80	14.3
A	degeneracy	C172	ContractRF	36	3.70	2.8
A	degeneracy	C172	Final	36	4.05	5.6
A	degeneracy	C172	Fixed2	36	3.70	2.8
A	degeneracy	C172	Fixed4	36	4.05	5.6
A	degeneracy	C172	FlatCART	36	3.78	2.8
A	degeneracy	C172	FlatRF	36	3.70	2.8
A	degeneracy	C172	KG1	36	3.78	2.8
A	degeneracy	C172	OldRAVC	36	3.72	2.8
A	map_shift	B767	ContractRF	29	7.01	20.7
A	map_shift	B767	Final	29	6.44	27.6
A	map_shift	B767	Fixed2	29	7.01	20.7
A	map_shift	B767	Fixed4	29	6.44	27.6
A	map_shift	B767	FlatCART	29	6.50	20.7
A	map_shift	B767	FlatRF	29	7.01	20.7
A	map_shift	B767	KG1	29	6.50	20.7
A	map_shift	B767	OldRAVC	29	6.80	20.7
A	map_shift	C172	ContractRF	35	11.05	82.9
A	map_shift	C172	Final	35	10.77	85.7
A	map_shift	C172	Fixed2	35	11.05	82.9
A	map_shift	C172	Fixed4	35	10.77	85.7
A	map_shift	C172	FlatCART	35	10.75	88.6
A	map_shift	C172	FlatRF	35	11.05	82.9
A	map_shift	C172	KG1	35	10.75	88.6
A	map_shift	C172	OldRAVC	35	11.11	85.7
A	occlusion	B767	ContractRF	35	5.63	28.6
A	occlusion	B767	Final	35	4.84	20.0
A	occlusion	B767	Fixed2	35	5.63	28.6
A	occlusion	B767	Fixed4	35	4.84	20.0
A	occlusion	B767	FlatCART	35	4.88	22.9
A	occlusion	B767	FlatRF	35	5.63	28.6
A	occlusion	B767	KG1	35	4.88	22.9

Study	Stress	Geometry	Policy	n	RMSE (cm)	Union (%)
A	occlusion	B767	OldRAVC	35	5.04	22.9
A	occlusion	C172	ContractRF	29	5.94	31.0
A	occlusion	C172	Final	29	5.68	31.0
A	occlusion	C172	Fixed2	29	5.94	31.0
A	occlusion	C172	Fixed4	29	5.68	31.0
A	occlusion	C172	FlatCART	29	5.92	31.0
A	occlusion	C172	FlatRF	29	5.94	31.0
A	occlusion	C172	KG1	29	5.92	31.0
A	occlusion	C172	OldRAVC	29	5.94	31.0
A	prior_shift	B767	ContractRF	30	7.16	26.7
A	prior_shift	B767	Final	30	4.33	6.7
A	prior_shift	B767	Fixed2	30	7.16	26.7
A	prior_shift	B767	Fixed4	30	4.33	6.7
A	prior_shift	B767	FlatCART	30	4.19	6.7
A	prior_shift	B767	FlatRF	30	7.16	26.7
A	prior_shift	B767	KG1	30	4.19	6.7
A	prior_shift	B767	OldRAVC	30	5.35	13.3
A	prior_shift	C172	ContractRF	34	7.10	29.4
A	prior_shift	C172	Final	34	5.55	14.7
A	prior_shift	C172	Fixed2	34	7.10	29.4
A	prior_shift	C172	Fixed4	34	5.55	14.7
A	prior_shift	C172	FlatCART	34	5.61	14.7
A	prior_shift	C172	FlatRF	34	7.10	29.4
A	prior_shift	C172	KG1	34	5.61	14.7
A	prior_shift	C172	OldRAVC	34	6.27	20.6
B	degeneracy	B767	ContractRF	31	9.20	32.3
B	degeneracy	B767	Final	31	7.36	22.6
B	degeneracy	B767	Fixed2	31	9.20	32.3
B	degeneracy	B767	Fixed4	31	7.36	22.6
B	degeneracy	B767	FlatCART	31	7.70	25.8
B	degeneracy	B767	FlatRF	31	9.20	32.3
B	degeneracy	B767	KG1	31	7.70	25.8
B	degeneracy	B767	OldRAVC	31	8.36	29.0
B	degeneracy	C172	ContractRF	33	4.38	6.1
B	degeneracy	C172	Final	33	4.11	9.1
B	degeneracy	C172	Fixed2	33	4.38	6.1
B	degeneracy	C172	Fixed4	33	4.11	9.1
B	degeneracy	C172	FlatCART	33	4.22	6.1
B	degeneracy	C172	FlatRF	33	4.38	6.1
B	degeneracy	C172	KG1	33	4.22	6.1
B	degeneracy	C172	OldRAVC	33	4.38	6.1
B	map_shift	B767	ContractRF	26	8.44	34.6
B	map_shift	B767	Final	26	7.87	30.8
B	map_shift	B767	Fixed2	26	8.44	34.6
B	map_shift	B767	Fixed4	26	7.87	30.8
B	map_shift	B767	FlatCART	26	7.83	30.8
B	map_shift	B767	FlatRF	26	8.44	34.6
B	map_shift	B767	KG1	26	7.83	30.8
B	map_shift	B767	OldRAVC	26	7.88	30.8
B	map_shift	C172	ContractRF	38	10.39	44.7
B	map_shift	C172	Final	38	10.06	52.6
B	map_shift	C172	Fixed2	38	10.39	44.7
B	map_shift	C172	Fixed4	38	10.06	52.6
B	map_shift	C172	FlatCART	38	10.04	52.6
B	map_shift	C172	FlatRF	38	10.39	44.7

Study	Stress	Geometry	Policy	n	RMSE (cm)	Union (%)
B	map_shift	C172	KG1	38	10.04	52.6
B	map_shift	C172	OldRAVC	38	10.07	44.7
B	occlusion	B767	ContractRF	34	6.48	35.3
B	occlusion	B767	Final	34	5.69	35.3
B	occlusion	B767	Fixed2	34	6.48	35.3
B	occlusion	B767	Fixed4	34	5.69	35.3
B	occlusion	B767	FlatCART	34	6.09	35.3
B	occlusion	B767	FlatRF	34	6.48	35.3
B	occlusion	B767	KG1	34	6.09	35.3
B	occlusion	B767	OldRAVC	34	5.96	35.3
B	occlusion	C172	ContractRF	30	4.72	16.7
B	occlusion	C172	Final	30	4.63	20.0
B	occlusion	C172	Fixed2	30	4.72	16.7
B	occlusion	C172	Fixed4	30	4.63	20.0
B	occlusion	C172	FlatCART	30	4.90	20.0
B	occlusion	C172	FlatRF	30	4.72	16.7
B	occlusion	C172	KG1	30	4.90	20.0
B	occlusion	C172	OldRAVC	30	4.72	16.7
B	prior_shift	B767	ContractRF	29	5.70	10.3
B	prior_shift	B767	Final	29	5.11	13.8
B	prior_shift	B767	Fixed2	29	5.70	10.3
B	prior_shift	B767	Fixed4	29	5.11	13.8
B	prior_shift	B767	FlatCART	29	5.12	10.3
B	prior_shift	B767	FlatRF	29	5.70	10.3
B	prior_shift	B767	KG1	29	5.12	10.3
B	prior_shift	B767	OldRAVC	29	5.23	6.9
B	prior_shift	C172	ContractRF	35	6.47	17.1
B	prior_shift	C172	Final	35	4.85	8.6
B	prior_shift	C172	Fixed2	35	6.47	17.1
B	prior_shift	C172	Fixed4	35	4.85	8.6
B	prior_shift	C172	FlatCART	35	4.95	8.6
B	prior_shift	C172	FlatRF	35	6.47	17.1
B	prior_shift	C172	KG1	35	4.95	8.6
B	prior_shift	C172	OldRAVC	35	5.35	5.7

S14. Additional diagnostic figures retained from the executed studies

These figures preserve the all-service, tail and geometry detail moved out of the main narrative. Their observations, sampling units and frozen statistical allocations are unchanged.



Figure S2: The five originally supplied simulation views, preserved as qualitative geometry and platform context. They are distinct from the independently generated episode and inspection-task samples.

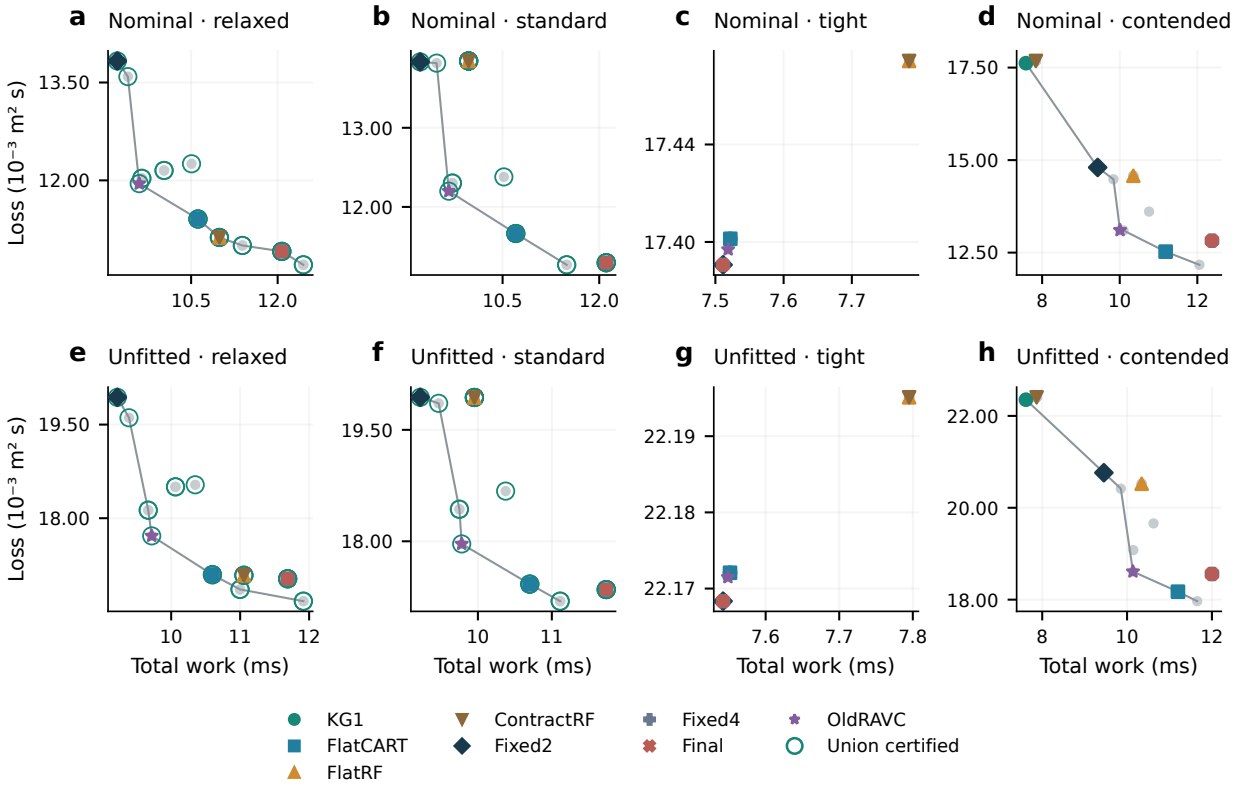


Figure S3: All 22 localization policy means across both confirmation populations and four services. Rings mark direct-union eligibility; Pareto lines are descriptive.

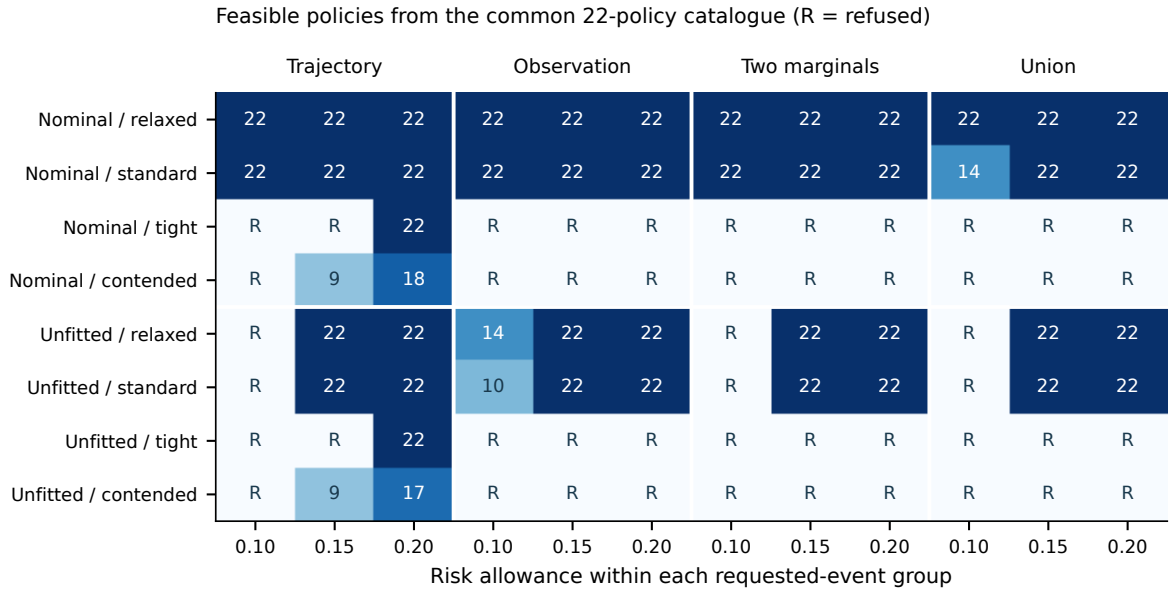


Figure S4: All 96 frozen localization requests across task and allowance variants. Empty catalogues are retained as refusals.

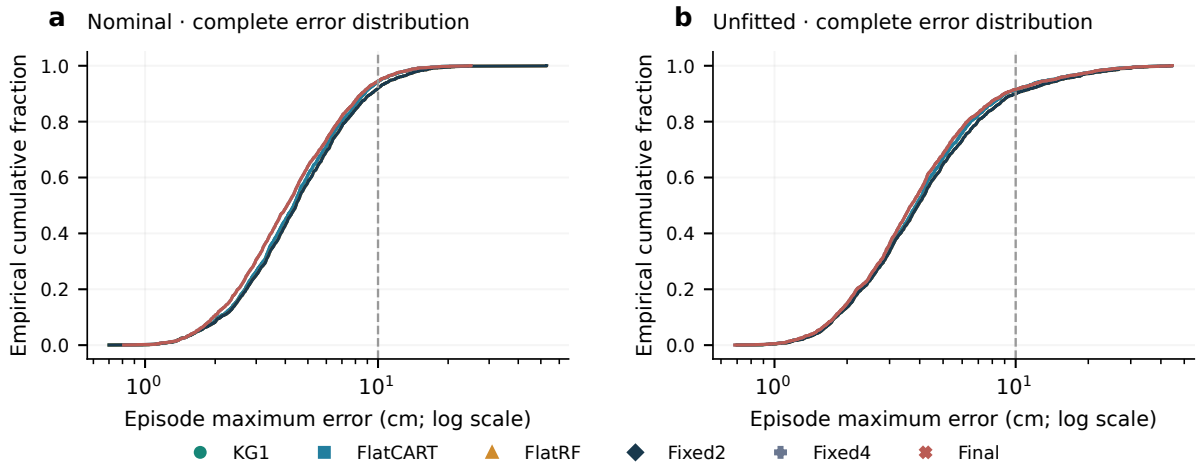


Figure S5: Empirical distributions of all confirmation episode maxima. The reference is 10 cm.

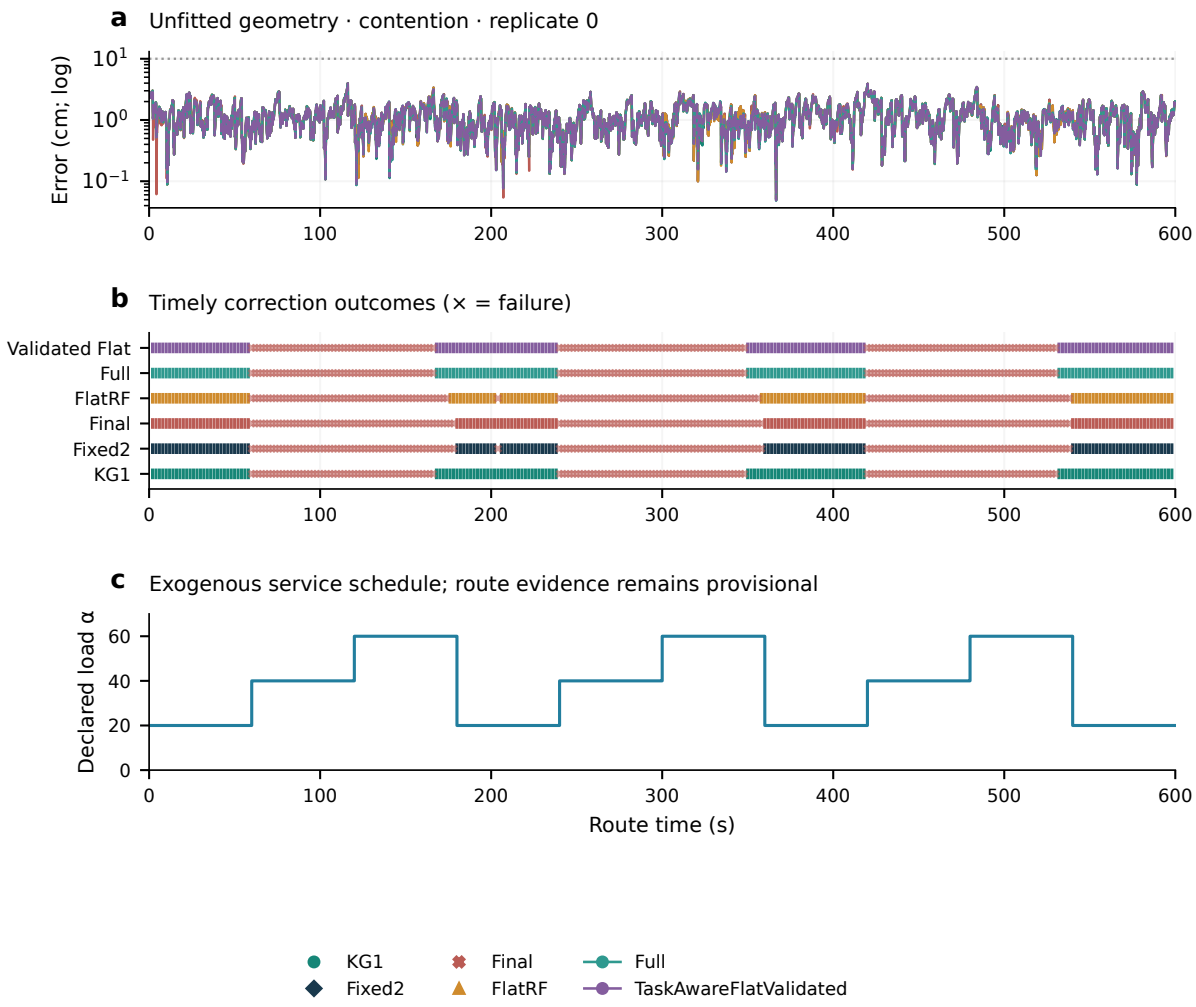


Figure S6: Predeclared unfitted-airframe/contention/replicate-zero route: all 6,000 position-error ticks, correction opportunities and exogenous processor load.

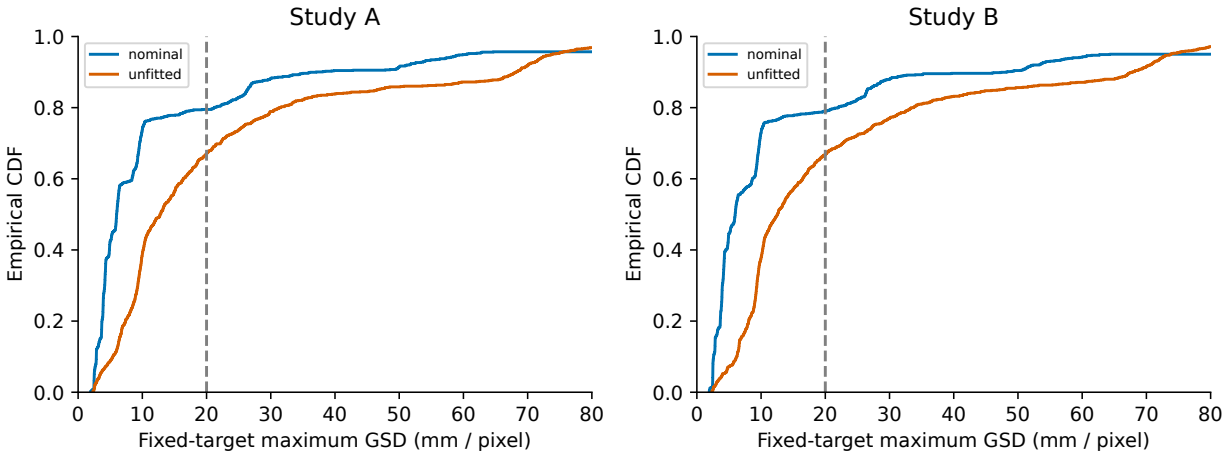


Figure S7: Fixed-target GSD empirical distributions on independent A/B confirmation samples. Planned targets outside the acquired ROI remain in the geometric score.

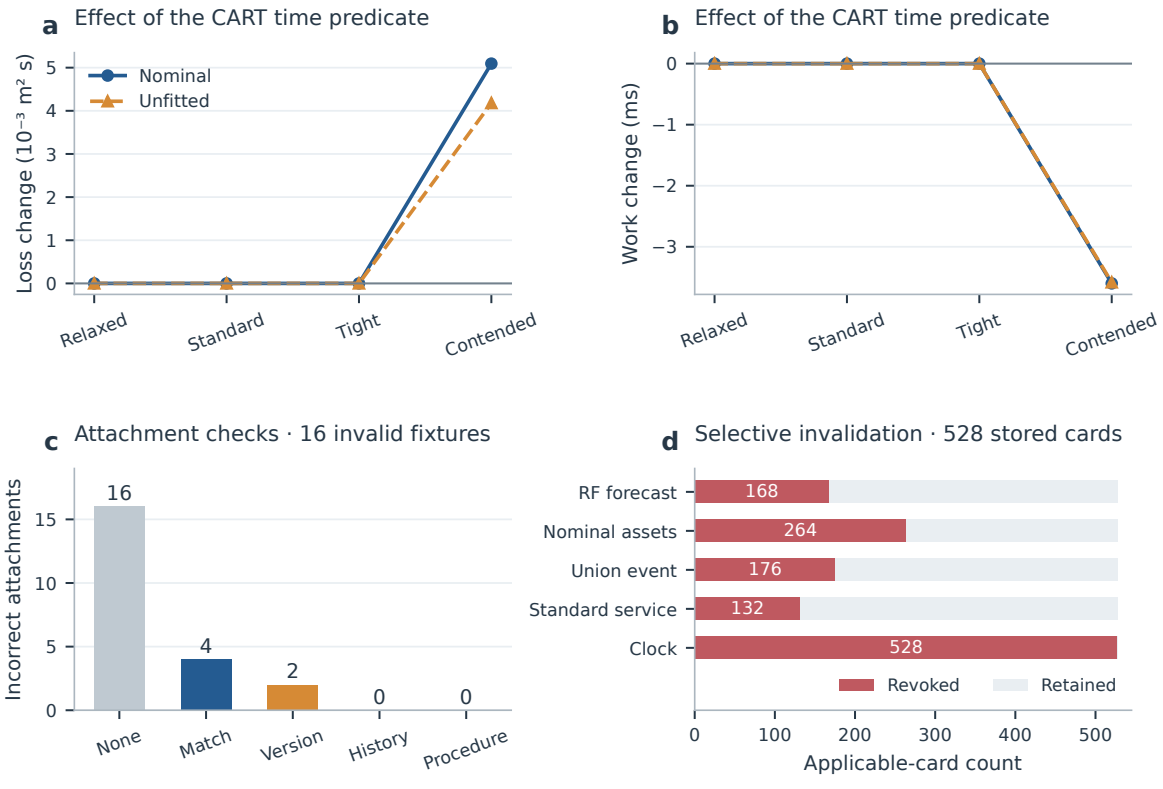


Figure S8: Localization mechanism diagnostics. Paired time-predicate effects across four services and two populations; incorrect attachments across 16 fixed invalid fixtures; and selective revocation among the original 528 localization cards. These counts retain their distinct original denominators.

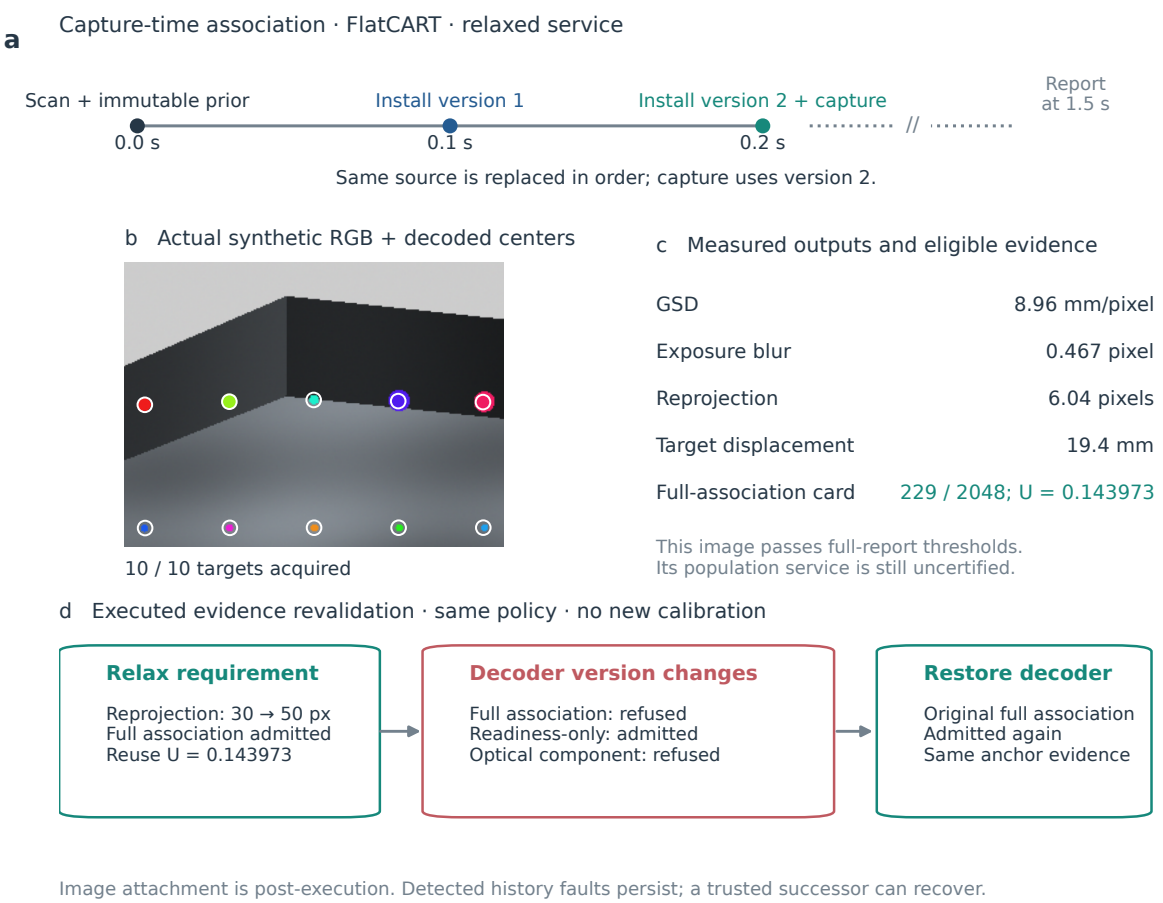


Figure S9: Original pixel-level diagnostic image, execution trace and evidence revalidation case. The raster is the actual archived synthetic output.

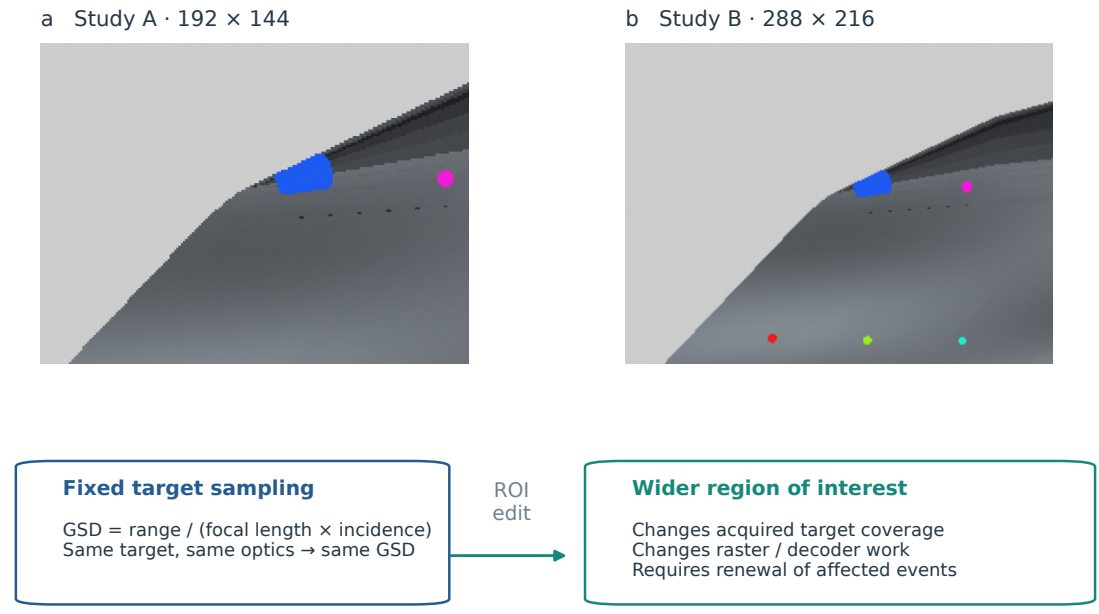
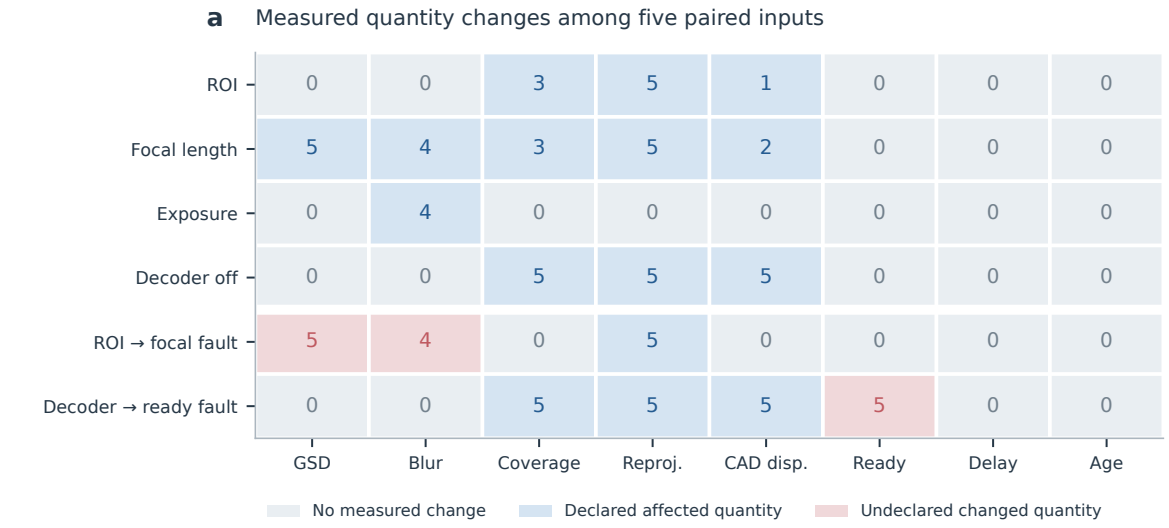


Figure S10: Original paired-ROI raster control with fixed CAD landmarks and optical parameters.



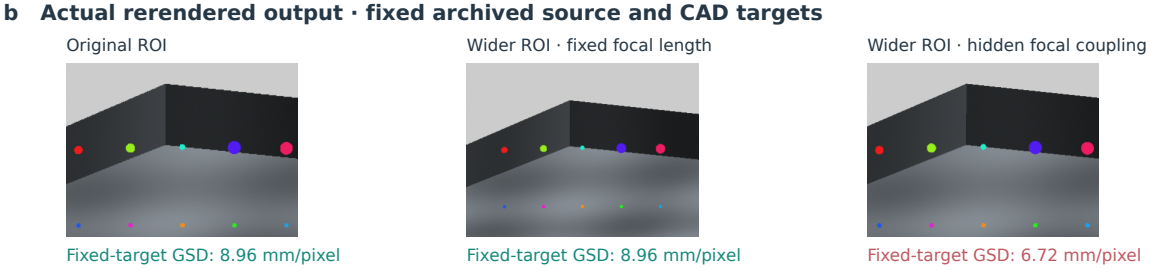


Figure S11: Original five-template producer re-execution, preceding the controlled camera campaign.

S15. Typed metamodel, compilation and executed producer auditing

The reference extension implements typed request compilation and a relation-domain checker. A request contains population, service, allowance and a conjunction of intervals. Each interval declares a metric and engineering unit, with optional lower and upper bounds. Unit conversion, interval intersection and canonical sorting produce the event identity; contradictory bounds and incompatible dimensions are rejected. All 24 declared task/variant expressions reproduce their original event identities and provider projections. Four invalid quantity specifications, a wrong relation type and an invalid source-revision order are rejected.

Table S38: Metamodel relation signatures implemented by the checker.

Relation	Domain	Range
requires	Request	Event
commits	Plan	Request
selects	Plan	Policy
bounds	Card	Event
supports	Card	Plan
derivedFrom	Card	Producer
produces	Producer	Quantity
dependsOn	Quantity	Provider
uses	Receipt	SourceVersion
records	Receipt	Plan
revises	SourceVersion	SourceVersion

For example, full association compiles coverage ≥ 0.50, reprojection ≤ 30 px, CAD-target displacement ≤ 100 mm and readiness ≥ 1. The compiler converts 100 mm to 0.10 m and gathers the declared dependencies of those quantities. The dependency manifest is an explicit implemented-data-flow specification; it is not inferred by a static-analysis engine. Unit/schema checks validate its use, while paired numerical re-execution audits its consequences.

The five sources are the previously sealed reproduction inputs: A/nominal 13,200,000 and 13,201,023; A/unfitted 13,300,000; B/nominal 14,200,000; and B/unfitted 14,300,000. For each, original CAD addresses, geometry perturbation, motion, capture time and installed FlatCART/relaxed state are retained. The renderer, pixel decoder and task scorer run on these fixed inputs. Baseline RGB arrays agree byte-for-byte with their archives and task quantities agree within 10^{-10}. Four interventions widen ROI by 96 by 72 pixels, set focal length to 2,400 px, set exposure to 9 ms, or disable the decoder. Two fault fixtures additionally make focal length depend on ROI width, or make capture readiness depend on the decoder's finite outputs, without declaring that coupling. The fixed virtual reporting clock is retained to isolate producer-function changes.

Measured quantity changes are computed from output values, independently of the dependency checker. A declared affected quantity may show no change on a finite input; this is not evidence of global invariance. An undeclared measured change falsifies the transfer premise for that tested producer/input. The replay audit flags all five pairs in each injected coupling class. These fixtures test the specified faults and do not estimate arbitrary hidden-coupling detection rates.

S15.1. Reuse-first routing and calibration-scope sensitivity

The routed evidence analysis uses all 320 unique primary contexts. It fixes the originally selected direct-renewal policy, selects a compatible covering anchor when available and otherwise retains an exact-event card. All applicable invariant evidence pools are available to both routes. Selected proofs are deduplicated by event, producer, population and calibration-source manifest; repeated service labels do not add distinct proofs. The catalogue counts measure selected proof-construction requirements, excluding acquisition, producer execution, policy-catalogue search, native timing and human effort.

Canonical calibration scopes use the same four signatures and do not merge tests using observed numerical coincidence. The anchor family contains 780 canonical tests; the complete stored family contains 1,692. Each

Table S39: Actual producer re-execution. Counts refer to five paired archived inputs per intervention.

Intervention	Pairs	Undeclared changes	Flagged pairs
roi	5	0	0
focal	5	0	0
exposure	5	0	0
decoder_disabled	5	0	0
hidden_roi_focal	5	9	5
hidden_decoder_readiness	5	5	5

Table S40: Selected proof preparation on identical supported contexts and fixed numerical policies.

Study	Route	Contexts	Proof cards	Record evaluations
A	Exact event	30	18	36864
A	Reuse + fallback	30	15	30720
B	Exact event	29	17	36864
B	Reuse + fallback	29	15	32768
combined	Exact event	59	35	73728
combined	Reuse + fallback	59	30	63488

sensitivity design separately allocates total confidence error 0.05. A combined routed catalogue requires one allocation over its tested union. The original primary cards continue to use 0.05/4,608. The canonical designs tighten the example bound without changing support at allowance 0.15.

Table S41: Archived-count scope sensitivity for the full-association example, with 229 failures among 2,048 calibration inputs.

Scope design	Method	Family size	Upper bound
Primary reserved	Anchor	4608	0.143973
Primary reserved	Direct	4608	0.143973
Canonical scope design	Anchor	780	0.140651
Canonical scope design	Direct	1692	0.142133

S15.2. Registration point-count probes

The unchanged source solver is run on the existing frontend cloud at requested caps 96, 192 and 384. Actual sample count clips to available points. Each run uses the original immutable prior; no forecast model is retrained. Native Linux CPU is the median of three repeated executions, including solver construction. The error column describes the final candidate before acceptance; rejected candidates remain visible. These finite probes use no primary budget wrapper or larger-point risk certificate.

Table S42: Paired point-count diagnostic on the five archived inputs. Timing repeats do not increase the source count.

Source	Cap	Used	Iterations	Accepted	Error (mm)	CPU (ms)
13200000	96	96	5	no	82.68	14.68
13200000	192	189	5	yes	88.23	16.01
13200000	384	189	5	yes	88.23	15.47
13201023	96	96	4	yes	81.79	8.71
13201023	192	192	6	yes	39.32	14.39
13201023	384	384	5	yes	73.40	14.40
13300000	96	96	3	yes	27.74	8.80
13300000	192	192	5	yes	11.29	11.83
13300000	384	205	5	yes	13.56	14.38
14200000	96	96	5	no	147.08	12.82
14200000	192	192	7	no	83.13	16.82
14200000	384	384	8	yes	87.18	17.85
14300000	96	96	3	yes	19.38	6.09
14300000	192	192	3	yes	16.04	7.69
14300000	384	384	3	yes	28.17	9.76

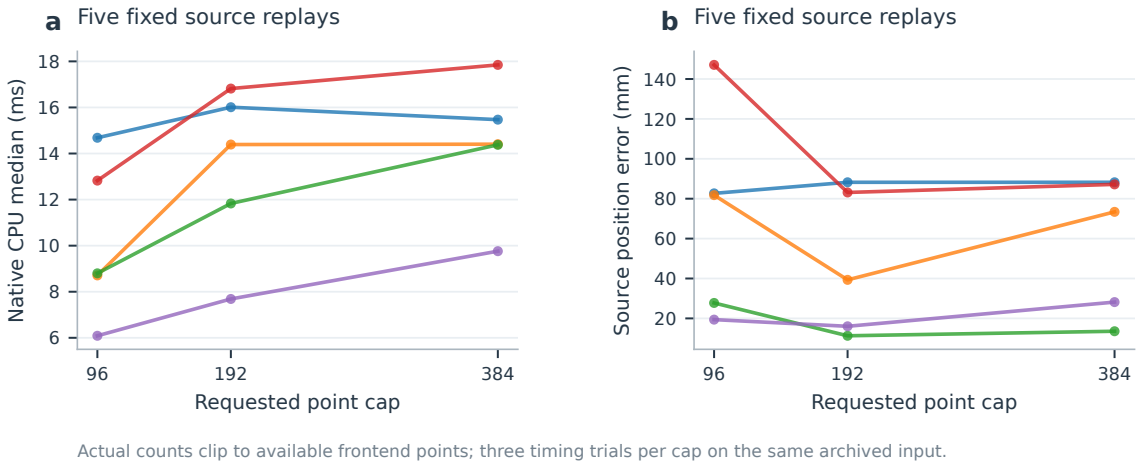


Figure S12: Executed point-count probes. Every trace is one fixed archived source. Candidate error is nonmonotonic in requested point count; timing medians describe this Linux diagnostic.

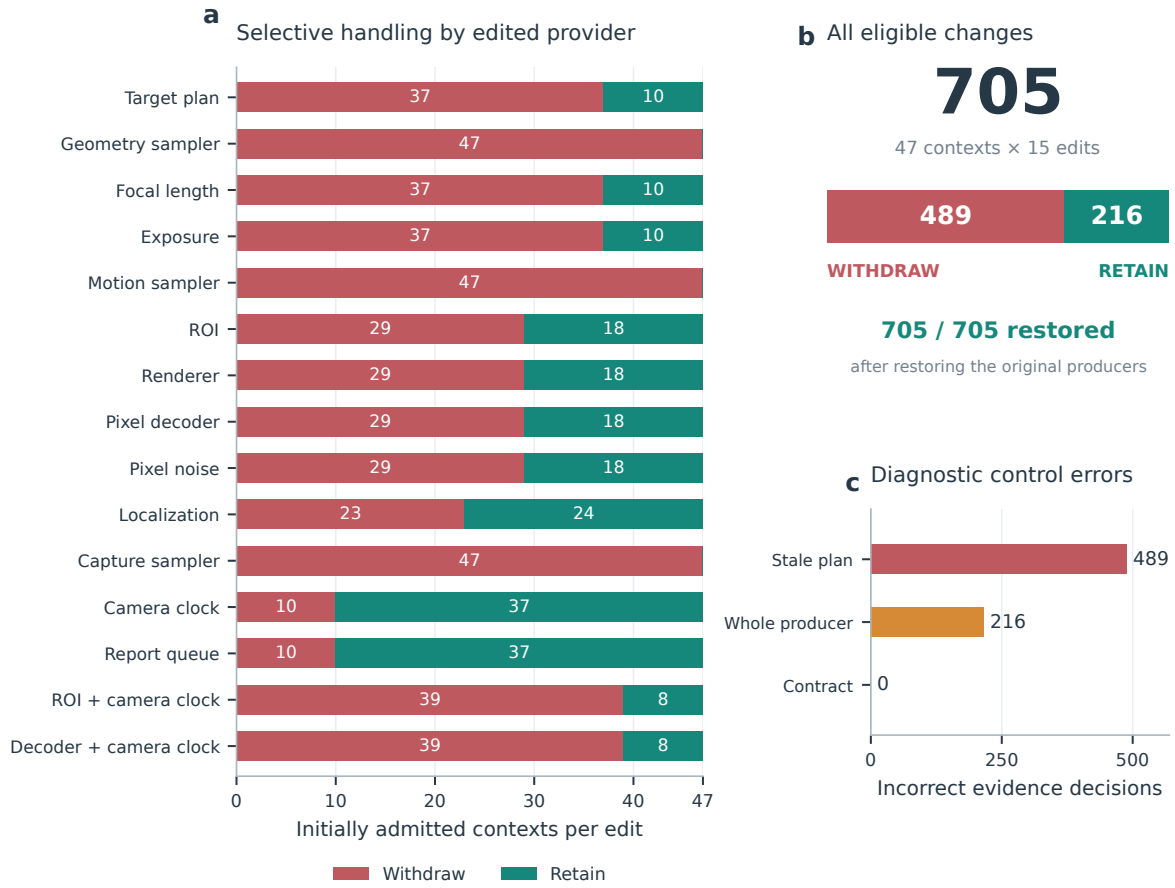


Figure S13: Original declared-model change/recovery conformance matrix, retained as implementation verification. The oracle shares the declaration specification; these counts do not test the correctness of an undeclared physical dependency.

S16. Controlled camera-change campaign

Five archived CAD/capture templates supply nominal A sources 13200000 and 13201023, unfitted A source 13300000, nominal B source 14200000 and unfitted B source 14300000. Each supplies 64 independently seeded camera perturbations. These are simulated configurations conditional on the templates; they add no population calibration cards. Camera translation and rotation have standard deviations 0.04 m and 0.001 rad. Focal length, exposure and brightness are uniform over 1600–2000 pixels, 4–8 ms and 0.70–1.15. Baseline and expanded ROIs are 384 by 288 and 512 by 384. Existing CAD addresses and installed localization estimates are retained.

The paired producer executes geometry, BVH raycasting, Gaussian exposure filtering, pixel decoding and task scoring. Focal-length updates multiply the original value by 1.20; exposure updates multiply it by 1.50; decoder updates suppress decoded output. Each pair uses the same noise seed. ROI–focal faults multiply focal length by 1.15 when the original focal length is at least 1800 pixels. Decoder–readiness faults make readiness depend on decoded output when exposure is at least 6 ms. Exposure–focal faults multiply focal length by 1.03 when edited exposure exceeds 10 ms. ROI–clock faults scale the camera operation clock with ROI area. Each faulty update advertises only its initiating provider change.

Table S43: Executed updates, conditional activation and regression vetoes.

Update	Pairs	Active faults	Vetoes	Undeclared changes
ROI expansion	320	0	0	0
Focal length	320	0	0	0
Exposure	320	0	0	0
Decoder off	320	0	0	0
ROI–focal coupling	320	163	163	288
Decoder–readiness coupling	320	160	160	160
Exposure–focal coupling	320	104	104	104
ROI–clock coupling	320	320	320	640

The configurations, all 20,480 paired quantity comparisons, change-case decisions and output summaries are supplied as CSV. The frozen script and input hashes permit standalone regeneration. Numerical verification compares recomputed values at tolerance 1e-10; CPU timing is not used as the detection criterion. Original primary risk cards and policy plans remain unchanged.

S17. Evidence scope, data provenance and declarations

Primary-study provenance and scope. The main lifecycle study independently samples 256 calibration scenes and 64 test scenes. Eight producer configurations execute on every calibration scene, producing 2,048 outputs for the fixed 64-event family. The test campaign visits six configurations and follows each source through 16 operations; paired stress inputs have a separate population identity. The surface challenge uses eight development scenes and 64 test scenes from four structural families, each under four imaging conditions. Its 256 test images are actual Blender Cycles outputs processed by the two fixed image providers. The latter two structural families are excluded from provider selection. These are model-generated observations, not physical aircraft measurements. Analytic geometry, sampled appearance, missingness, occlusion and prescribed CAD/prior errors define the evaluated populations. Calibration records, test outcomes and image-processing parameters remain attached to their respective populations.

Figure 1 combines two original project photographs with two openly reusable photographs of aircraft skin. The actual-input figures show controlled raster outputs and public real-image defect inputs with deterministic annotation/prediction overlays; supplementary galleries also show Cycles outputs. The remaining figures report executed numerical results or explicit workflows. All image examples follow fixed source-order selection, and all test successes and failures remain in the numerical archive. Diagrams and quantitative graphics are produced by deterministic figure scripts. No generative scene artwork is included in either PDF. The four photographs in Figure 1 provide engineering context; the separate Innovation Hangar images are actual inputs to the defect-detection experiment.

Real-image dataset provenance, license and measurement scope. The real-image case uses Innovation Hangar v2, released version 1, from the public repository accompanying Suvittawat et al., *Aerospace* 12(4), 356 (2025), DOI 10.3390/aerospace12040356. Credit belongs to the Innovation Hangar dataset contributors and the source-study authors. The original project is `https://universe.roboflow.com/innovation-hangar/innovation-hangar-v2`; both its release and the mirrored export identify Creative Commons Attribution 4.0. Upstream source-photo credits remain applicable. The archive's `SOURCE_CREDIT.md` records the dataset, software and model licenses separately.

The released 10,722 images reduce to 1,339 representatives in 222 inferred source groups after filename-family selection and visual merges. These groups are inferred from source identifiers and capture dates; aircraft or full acquisition identities are unavailable. Statistical bounds assume independent complete source-group draws and a matched population. Grouping checks reduce known repeat dependence but do not prove all remaining groups independent. The three less represented categories each occupy only one calibration/test source group. Their per-frame results remain descriptive, and pooled event cards do not certify per-category inspection quotas. Digital gain variants transform existing images; they do not establish newly measured daylight, night or illumination conditions.

Development selects the detector weights and operational threshold before calibration/test prediction. Every representative image, original and converted annotation, selection record, model weight, raw prediction and event outcome is archived. This source-group partition differs from the source paper's released image split; performance figures from that paper are not transferred to this detector. Repeated calibration subsets are descriptive stability checks on the same data. Model training and inference have separate measured times, distinct from the earlier CAD workflow's computation timer.

Open photograph credits and reuse. Figure 1c: Simon Butler, "655 Maintenance and Preservation Society ex RAF Avro Vulcan B.2A XM655/G-VULC under fuselage detail," 17 September 2023. Licensed under Creative Commons Attribution 2.0. Source: original Flickr photograph, via Wikimedia Commons. The image is scaled for the figure without retouching. Copyright remains with the photographer.

Figure 1d: TSGT Rob Marshall, "A rivet is inserted in a sheet metal aircraft patch during Exercise NIGHT TRAIN/GLOBAL SHIELD '84," Davis–Monthan Air Force Base, 2 April 1984. U.S. Department of Defense photograph, public domain; National Archives identifier 6390063. Source: U.S. National Archives, via Wikimedia Commons and the Digital Public Library of America. The image is scaled without retouching.

Both photographs illustrate engineering context. Neither supplies observations, labels or calibration records for the quantitative benchmark. The editable package preserves the downloaded originals, source-page URLs, author and license metadata, acquisition dates and SHA-256 checksums in `paper/figures/open_photos/sources.json`.

Complementary-study provenance and attribution. The earlier four-layout study retains its 256 calibration scenes, 256 test scenes, six producers and 48-event allocation, together with 128 stress inputs, 16 timing inputs and 64 component-regression inputs. Its 24-panel Cycles transfer study retains all 72 images. These records are distinct from the new 64-event calibration and four-structure challenge. The numerical localization, route, diagnostic-fiducial image and conditional camera-update studies likewise retain their original identities, clocks and allocations. The 320 camera configurations are perturbations of five CAD/capture templates. The original five-source audit reproduces archived baseline pixels and quantities; specified conditional coupling fixtures test numerical regression, while the 705-edit matrix checks the declared rules. The 20261014 regular-grid study retains its 384 scenes and 48-event allocation. None of these earlier measurements is pooled into the new calibration family.

In the retained standard-service localization results, Contract-CART and FlatCART share numerical gains; Fixed4/Final and FlatRF/ContractRF have coincident reported means. Time-only CART also coincides at relaxed/standard service. The profile predicates change none of those test outputs because ordinary priors and declared noise remain below their first thresholds and the unfitted reservation limits refinement. Time-screen effects occur under contention. These observations do not establish global policy identity; the original 22-policy family and allocation remain fixed.

Full and the complete procedure reproduce numerical and execution-record outcomes, including 18/32 localization admissions. Contract and complete procedural coverage agree across 6,144 diagnostic image-task plans, test decisions and declaration-model conformance cases. The earlier 48-event comparison reduces record evaluations by 38.10%. The current complete 64-event catalogue reduces them by 35.19%, and the separate 16-step demand reduces them by 25%, each relative to exact dependency caching with identical support and confidence allocation. These are separate demand experiments, and each population catalogue is built once. Independent rules have the same event semantics and counts. The larger numerical computation saving is shared by the three DAG-based methods.

Statistical interpretation and regression scope. Population bounds assume independent complete calibration inputs, a fixed effective producer/event family, a matched population and trusted dependencies. Proposition 1 of the main article bounds the fresh-scene risk of calibration-selected services. The joint-failure statement bounds failure together with issuance. Receipt-conditioned, subgroup and route risks require their own calibration statements. Transfer requires requested-function identity throughout the supported input domain. Paired regression can falsify a proposed invariance by exposing an undeclared difference; finite successful examples do not prove global function identity. The regression gate runs when a component is registered, while runtime attachment checks registered signatures, numerical versions and authenticated binding state. The current recovery model retains invalid-attempt history and restores a binding through a validated transaction; the earlier permanent latch is preserved as a comparison.

The calibration/test split excludes development inputs. All physical quality labels are evaluated after pixel matching and estimation. Real-image defect annotations likewise enter scoring after the detector has produced its boxes from pixels. The current timing comparison pairs complete 16-step trajectories on 64 independent sources, with rotated method order and identical cache access for the three incremental methods. Its 1,024 operations per method are repeated workflow decisions, not independent image samples. The earlier 640-case source-renewal comparison, 81,920 permanent-latch decisions and 53,248 repair rows retain their own reported units. Catalogue construction separately measures event evaluation, bound inversion and routing over stored calibration quantities. Current transactions authenticate actual image/correspondence hashes, observable residuals and counts; they do not read physical mapping or correct-match labels. The authority is a controlled in-process registry. Hashes check integrity within that trust boundary; a compromised authority or distributed security protocol is outside the evaluated model.

Physical implementation and measurement scope. Physical implementation requires representative image observations, surface appearance and camera calibration, platform timing and population-matched risk calibration. The controlled model includes prescribed CAD discrepancies, occlusion and missing sites. Cycles adds geometric fasteners, reflectance and area illumination on synthetic surface CAD. The image-quality requests and allowance 0.15 are benchmark operating points inherited before sampling; they do not specify defect-detection, missed-defect or airworthiness probabilities. Lens distortion, deforming airframes and defect recognition remain outside the CAD-association service. The separate real-image service evaluates defect boxes against public annotations, with its own grouped events and producer family. The legacy virtual operation clock, Windows component CPU, whole-route CPU and record lookup times have separate boundaries. Linux point-count medians use three repetitions per archived input. The Linux source-renewal timings include requested numerical stages and offline quality scoring. In the current sequential campaign, event timing also includes payload hashing, registry and consistency checks, and the stress-source draw within the population-change event. Initial nominal-source sampling, process startup and file export lie outside that timer; catalogue construction is measured separately. The archived protocol's broader exclusion of source sampling applies to the initial nominal draw, while the executed stress draw remains inside the measured event. These are computational-workload timings, not online camera-to-decision latency claims.

Declaration of generative AI and AI-assisted technologies. OpenAI ChatGPT/Codex assisted code development, computational analysis, manuscript revision and figure scripts. Quantitative observations come from the archived or independently executed numerical producers. All manuscript images are original or credited open photographs, computed raster output or deterministic graphics; no generative image output supplies experimental evidence.

Funding. This work was supported by the Civil Aviation Safety Capacity Building Fund Project (RJ2025073), "Pilot-scale development of civil aircraft walk-around inspection and on-wing aero-engine cleaning equipment."

Data and code availability. The repository https://github.com/edrisskrah131-lang/A20-AEI-R8 provides the main article, supplement, editable source archive and associated data volumes. They contain all Cycles image inputs, exact surface meshes, sampled configurations, numerical observations, risk cards, transaction traces and figure recipes. The real-image case adds representative source pixels, original and converted annotations, pretrained/fine-tuned detector weights, all 3,524 candidate-detection arrays and their scoring records. Controlled-image studies retain representative raw arrays together with the seeds, numerical records and pixel hashes needed to reconstruct the remaining outputs. SHA-256 manifests identify every source and data file. The restoration utility verifies the data volumes before placing their files in the source tree. Reproduction scripts re-execute numerical producers, reprocess rendered images, check recovery invariants, regenerate figures and compile both PDFs. The documented Blender command reconstructs any challenge scene, and fixed-source fresh-render checks verify the archived pixels. Compiled article PDFs are downloaded separately. Earlier full population assets remain in their original versioned release.